Carl Friedrich Gauss

Arithmetical Papers

English translation

Volume II: first three papers and fourth paper, Articles 1–19

This selection contains the historical title matter and first three complete papers of Volume II: A New Proof of an Arithmetical Theorem (Articles 1–7), presented on January 15, 1808; Summation of Certain Special Series (Articles 1–36), presented on August 24, 1808; and New Proofs and Extensions of the Fundamental Theorem in the Theory of Quadratic Residues, presented on February 10, 1817. The third paper contains the introduction, fifth proof (Articles 1–3), sixth proof (Articles 1–6) and new algorithm (Articles 1–6). This selection also contains the title and Articles 1–19 of Theory of Biquadratic Residues, First Paper, presented on April 5, 1825. The remainder of that paper and the later papers of Volume II lie beyond this selection. Volume I accompanies it in the collected reader.

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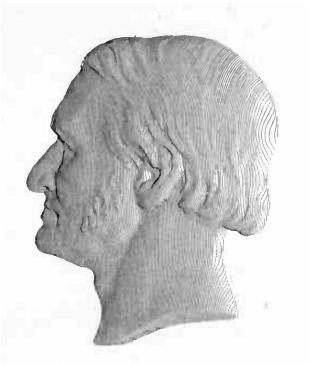
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A NEW PROOF

OF AN ARITHMETICAL THEOREM

BY

CARL FRIEDRICH GAUSS

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Commentationes societatis regiae scientiarum Gottingensis. Vol. XVI.
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A NEW PROOF

OF AN ARITHMETICAL THEOREM.

1.

Questions in higher arithmetic very often present a singular phenomenon, which occurs much more rarely in analysis and contributes greatly to their charm. In analytic investigations we usually cannot reach new truths unless we have first thoroughly mastered the principles on which they rest and which must, as it were, open the way to them. In arithmetic, by contrast, most elegant new truths very frequently spring forth through induction and a kind of unexpected good fortune; their proofs lie so deeply hidden and are wrapped in such darkness that they elude every attempt and deny access to the keenest investigations. Moreover, so great and so wonderful is the connection among arithmetical truths that at first sight seem utterly different, that quite often, while we are seeking something quite different, we at last most happily reach the proof so greatly desired and previously sought in vain through long reflection, by a path far different from the one expected. Usually, however, truths of this kind are of such a nature that they can be approached by several very different paths, and the paths that first present themselves are not always the shortest. It must therefore certainly be highly valued if, after such a truth has long been investigated in vain, and has then indeed been proved, but by rather obscure detours, we finally succeed in discovering the simplest and genuine path.

2.

Among the questions discussed in the preceding article, a distinguished place is held by the theorem containing almost the whole theory of quadratic residues, which in the *Arithmetical Investigations* (Section IV) is distinguished by the name of the *fundamental theorem*. The illustrious LEGENDRE must undoubtedly be regarded as the *first* discoverer of this most elegant theorem, after the great geometers EULER and LAGRANGE had, long before, already discovered several of its special cases by induction. I do not pause here to enumerate these men's attempts at a proof; those who wish may consult the work just mentioned. Let me add only, in confirmation of what was said in the preceding article, what concerns my own attempts. I came upon the theorem itself independently in the year 1795, when I was entirely ignorant of everything that had already been worked out in higher arithmetic and wholly cut off from access to the literature. But for an entire year it tormented me and eluded my most strenuous efforts, until at last I obtained the proof given in the fourth section of that work. Afterwards three others, resting on wholly

different principles, presented themselves to me; I gave one in the fifth section, and the others, not inferior to it in elegance, I shall make public on another occasion. But all these proofs, although they seem to leave nothing to be desired in rigour, are derived from principles that are too different in kind, perhaps excepting the first, which nevertheless proceeds by more laborious reasoning and is burdened by rather lengthy operations. I therefore do not hesitate to declare that a *genuine* proof has so far been lacking. Let the experts now judge whether the one that I recently succeeded in discovering, and that the following pages present, deserves to be honoured with this name.

3.

THEOREM. *Let p be a positive prime number; k any integer not divisible by p; use the following definitions:*

A *collection of the numbers* $1,\ 2,\ 3\ \ldots.\ \frac{1}{2}(p-1)$

B *collection of these* $\frac{1}{2}(p+1),\ \frac{1}{2}(p+3),\ \frac{1}{2}(p+5)\ \ldots.\ p-1$

Take the products of k by the individual numbers in A, and their least positive residues modulo p, which are evidently all distinct and belong partly to A, partly to B. Now if altogether B contains μ of the residues, k will be either a quadratic residue or a quadratic nonresidue of p, according as μ is even or odd.

Proof. Let the residues belonging to A be $a,\ a'\ a''\ \ldots.$, and the remaining ones, belonging to B, be $b, b', b'' \ldots.$. It is clear that the complements of the latter, $p-b, p-b', p-b'' \ldots.$, are all distinct from the numbers $a,\ a',\ a''\ \ldots.$, but together with these fill out the collection A. Thus we have

$$1\,.\,2\,.\,3\ \ldots.\ \tfrac{1}{2}(p-1) = aa'a''\ \ldots.\ (p-b)(p-b')(p-b'')\ \ldots.$$

But the latter product evidently becomes

$$\equiv (-1)^{\mu} aa'a''\ \ldots.\ bb'b''\ \ldots.\ \equiv (-1)^{\mu} k\,.\,2k\,.\,3k\ \ldots.\ \tfrac{1}{2}(p-1)k$$
$$\equiv (-1)^{\mu} k^{\frac{1}{2}(p-1)}\,1\,.\,2\,.\,3\ \ldots.\ \tfrac{1}{2}(p-1) (\mathrm{mod}.p)$$

Hence we have

$$1 \equiv (-1)^{\mu} k^{\frac{1}{2}(p-1)}$$

or $k^{\frac{1}{2}(p-1)} \equiv \pm 1$, according as μ is even or odd; our theorem follows immediately.

4.

We can greatly shorten the following reasoning by introducing some suitable notation. Thus the symbol (k, p) will denote for us the number of products among

$$k,\ 2k,\ 3k\ \ldots.\ \tfrac{1}{2}(p-1)k,$$

whose least positive residues modulo p exceed half of it. Further, when x is any noninteger quantity, we shall use the symbol $[x]$ to denote the integer immediately smaller than x itself, so that $x - [x]$ is always a positive quantity lying between the limits 0 and 1. The following relations are now obtained with little trouble:

I. $[x] + [-x] = -1$.

II. $[x] + h = [x + h]$, whenever h is an integer.

III. $[x] + [h - x] = h - 1$.

IV. If $x - [x]$ is a fraction smaller than $\frac{1}{2}$, then $[2x] - 2[x] = 0$; but if $x - [x]$ is greater than $\frac{1}{2}$, then $[2x] - 2[x] = 1$.

V. Thus, when the least positive residue of the integer h modulo p lies below $\frac{1}{2}p$, we have $\left[\frac{2h}{p}\right] - 2\left[\frac{h}{p}\right] = 0$; but when that residue lies above $\frac{1}{2}p$, we have $\left[\frac{2h}{p}\right] - 2\left[\frac{h}{p}\right] = 1$.

VI. It follows at once that $(k, p) =$

$$\left[\frac{2k}{p}\right] + \left[\frac{4k}{p}\right] + \left[\frac{6k}{p}\right] \ldots. + \left[\frac{(p-1)k}{p}\right]$$
$$- 2\left[\frac{k}{p}\right] - 2\left[\frac{2k}{p}\right] - 2\left[\frac{3k}{p}\right] \ldots. - 2\left[\frac{\frac{1}{2}(p-1)k}{p}\right].$$

VII. From VI and I we obtain without difficulty

$$(k, p) + (-k, p) = \tfrac{1}{2}(p-1)$$

It follows that $-k$ has either the same or the opposite relation to p (insofar as it is a quadratic residue or nonresidue of it) as $+k$, according as p is of the form $4n + 1$ or of the form $4n + 3$. In the former case -1 is evidently a residue, in the latter a nonresidue, of p.

VIII. We shall transform the formula given in VI as follows. By III we have

$$\left[\frac{(p-1)k}{p}\right] = k - 1 - \left[\frac{k}{p}\right],$$
$$\left[\frac{(p-3)k}{p}\right] = k - 1 - \left[\frac{3k}{p}\right],$$
$$\left[\frac{(p-5)k}{p}\right] = k - 1 - \left[\frac{5k}{p}\right] \ldots.$$

Applying these substitutions to the last $\frac{p\mp 1}{4}$ terms of the upper series in that expression, we have

first, whenever p is of the form $4n+1$

$$(k,p) = \tfrac{1}{4}(k-1)(p-1)$$
$$- 2\left\{\left[\frac{k}{p}\right] + \left[\frac{3k}{p}\right] + \left[\frac{5k}{p}\right] \ldots. + \left[\frac{\frac{1}{2}(p-3)k}{p}\right]\right\}$$
$$- \left\{\left[\frac{k}{p}\right] + \left[\frac{2k}{p}\right] + \left[\frac{3k}{p}\right] \ldots. + \left[\frac{\frac{1}{2}(p-1)k}{p}\right]\right\}$$

second, whenever p is of the form $4n+3$

$$(k,p) = \tfrac{1}{4}(k-1)(p+1)$$
$$- 2\left\{\left[\frac{k}{p}\right] + \left[\frac{3k}{p}\right] + \left[\frac{5k}{p}\right] \ldots. + \left[\frac{\frac{1}{2}(p-1)k}{p}\right]\right\}$$
$$- \left\{\left[\frac{k}{p}\right] + \left[\frac{2k}{p}\right] + \left[\frac{3k}{p}\right] \ldots. + \left[\frac{\frac{1}{2}(p-1)k}{p}\right]\right\}$$

IX. For the special case $k = +2$, the formulas just given yield $(2,p) = \frac{1}{4}(p\mp 1)$, taking the upper or lower sign according as p is of the form $4n+1$ or $4n+3$. Thus $(2,p)$ is even, and hence $2Rp$, whenever p is of the form $8n+1$ or $8n+7$; on the other hand, $(2,p)$ is odd and $2Np$, whenever p is of the form $8n+3$ or $8n+5$.

5.

THEOREM. *Let x be a positive noninteger quantity, among whose multiples x, $2x$, $3x \ldots$. up to nx none is an integer; put $[nx] = h$, from which it is easily concluded that among the multiples of the reciprocal quantity $\frac{1}{x}$, $\frac{2}{x}$, $\frac{3}{x} \ldots\ldots$ up to $\frac{h}{x}$ an integer is likewise not found. Then I say that we have*

$$\left.\begin{array}{l} [x] + [2x] + [3x] \,.\, + [nx] \\ + \left[\frac{1}{x}\right] + \left[\frac{2}{x}\right] + \left[\frac{3}{x}\right] \ldots. + \left[\frac{h}{x}\right] \end{array}\right\} = nh.$$

Proof. In the series $[x] + [2x] + [3x] \ldots. + [nx]$, which we shall put $= \Omega$, the first terms up to and including the term in position $\left[\frac{1}{x}\right]$ are evidently all $= 0$; the following ones up to the term in position $\left[\frac{2}{x}\right]$ are all $= 1$; the following ones up to the term in position $\left[\frac{3}{x}\right]$ are all $= 2$, and so on. Hence we have

$$\left.\begin{array}{l} \Omega = 0 \times \left[\frac{1}{x}\right] \\ \quad + 1 \times \left\{\left[\frac{2}{x}\right] - \left[\frac{1}{x}\right]\right\} \\ \quad + 2 \times \left\{\left[\frac{3}{x}\right] - \left[\frac{2}{x}\right]\right\} \\ \quad + 3 \times \left\{\left[\frac{4}{x}\right] - \left[\frac{3}{x}\right]\right\} \\ \qquad \text{etc.} \\ \quad + (h-1) \left\{\left[\frac{h}{x}\right] - \left[\frac{h-1}{x}\right]\right\} \\ \quad + h \left\{n - \left[\frac{h}{x}\right]\right\} \end{array}\right\} = hn - \left[\frac{1}{x}\right] - \left[\frac{2}{x}\right] - \left[\frac{3}{x}\right] \ldots. - \left[\frac{h}{x}\right]$$

Q. E. D.

6.

THEOREM. *If k, p denote any positive odd numbers relatively prime to each other, we have*

$$\left.\begin{aligned}&\left[\frac{k}{p}\right]+\left[\frac{2k}{p}\right]+\left[\frac{3k}{p}\right]\ldots.+\left[\frac{\frac{1}{2}(p-1)k}{p}\right]\\&+\left[\frac{p}{k}\right]+\left[\frac{2p}{k}\right]+\left[\frac{3p}{k}\right]\ldots.+\left[\frac{\frac{1}{2}(k-1)p}{k}\right]\end{aligned}\right\}=\tfrac{1}{4}(k-1)(p-1).$$

Proof. Supposing, as we may, that $k<p$, we have $\frac{\frac{1}{2}(p-1)k}{p}$ less than $\frac{1}{2}k$, but greater than $\frac{1}{2}(k-1)$, and therefore $\left[\frac{\frac{1}{2}(p-1)k}{p}\right]=\frac{1}{2}(k-1)$. Thus the present theorem evidently follows immediately from the preceding one by putting there $\frac{k}{p}=x$, $\frac{1}{2}(p-1)=n$, and hence $\frac{1}{2}(k-1)=h$.

Moreover, it can be proved in a similar manner that, if k is an *even* number relatively prime to p, we have

$$\left.\begin{aligned}&\left[\frac{k}{p}\right]+\left[\frac{2k}{p}\right]+\left[\frac{3k}{p}\right]\ldots.+\left[\frac{\frac{1}{2}(p-1)k}{p}\right]\\&+\left[\frac{p}{k}\right]+\left[\frac{2p}{k}\right]+\left[\frac{3p}{k}\right]\ldots.+\left[\frac{\frac{1}{2}kp}{k}\right]\end{aligned}\right\}=\tfrac{1}{4}k(p-1)$$

But we do not dwell on this proposition, which is not needed for our purpose.

7.

The fundamental theorem now follows immediately by combining the preceding theorem with proposition VIII of Article 4. Namely, if k, p denote any unequal positive prime numbers, and we put

$$(k,p) + \left[\frac{k}{p}\right] + \left[\frac{2k}{p}\right] + \left[\frac{3k}{p}\right] \ldots . + \left[\frac{\frac{1}{2}(p-1)k}{p}\right] = L$$

$$(p,k) + \left[\frac{p}{k}\right] + \left[\frac{2p}{k}\right] + \left[\frac{3p}{k}\right] \ldots . + \left[\frac{\frac{1}{2}(k-1)p}{k}\right] = M$$

it is clear from VIII of Article 4 that L and M are always even numbers. But by the theorem of Article 6 we have

$$L + M = (k,p) + (p,k) + \tfrac{1}{4}(k-1)(p-1)$$

Thus whenever $\frac{1}{4}(k-1)(p-1)$ is even, which happens if either both k, p or at least one of them is of the form $4n+1$, (k,p) and (p,k) must necessarily be either both even or both odd. But whenever $\frac{1}{4}(k-1)(p-1)$ is odd, which happens if both k, p are of the form $4n+3$, one of the numbers (k,p), (p,k) must necessarily be even and the other odd. Thus in the former case the relation of k to p and the relation of p to k (insofar as one is a residue or nonresidue of the other) are identical, and in the latter case opposite.

Q. E. D.

SUMMATION

OF CERTAIN

SPECIAL SERIES

BY

CARL FRIEDRICH GAUSS

PRESENTED TO THE SOCIETY AUGUST XXIV, MDCCCVIII.

Commentationes societatis regiae scientiarum Gottingensis recentiores. Vol. I.
Gottingae MDCCCXI.

SUMMATION

OF CERTAIN SPECIAL SERIES.

1.

Among the more notable truths to which the theory of circle division has opened the way, the summation proposed in Disquiss. Arithmet., Article 356, holds a distinguished place. This is due both to its particular elegance and remarkable fruitfulness, which another investigation will later give us occasion to discuss more fully, and to the unusual difficulties that beset its rigorous and complete proof. These difficulties should have been all the less expected because they concern a certain restriction of the theorem more than the theorem itself: once this restriction is disregarded, a proof is immediately available and follows very easily from the theory developed in that work. The theorem was given there in the following form. Suppose that n is a prime number. Represent all the quadratic residues of n between 1 and $n-1$ inclusive by a, and all the nonresidues between the same limits by b; finally, let ω be the arc $\frac{360^0}{n}$, and k any fixed integer not divisible by n. Then we have

I. for a value of n of the form $4m+1$,

$$\begin{aligned}
\Sigma \cos ak\omega &= -\tfrac{1}{2} \pm \tfrac{1}{2}\sqrt{n} \\
\Sigma \cos bk\omega &= -\tfrac{1}{2} \mp \tfrac{1}{2}\sqrt{n}, \text{ and hence} \\
\Sigma \cos ak\omega - \Sigma \cos bk\omega &= \pm\sqrt{n} \\
\Sigma \sin ak\omega &= 0 \\
\Sigma \sin bk\omega &= 0
\end{aligned}$$

II. for a value of n of the form $4m+3$,

$$\begin{aligned}
\Sigma \cos ak\omega &= -\tfrac{1}{2} \\
\Sigma \cos bk\omega &= -\tfrac{1}{2} \\
\Sigma \sin ak\omega &= \pm\tfrac{1}{2}\sqrt{n} \\
\Sigma \sin bk\omega &= \mp\tfrac{1}{2}\sqrt{n} \\
\Sigma \sin ak\omega - \Sigma \sin bk\omega &= \pm\sqrt{n}
\end{aligned}$$

These summations were proved with full rigor in the place cited, and the only difficulty that remains here is to determine the *sign* to be placed before the radical. It can indeed be shown without difficulty that this sign depends on the number k to the following extent: the *same* sign must hold for all values of k that are quadratic residues of n, and the opposite sign for all values of k

that are quadratic nonresidues of n. Thus the whole question turns on the value $k = 1$: as soon as the sign for this value is known, the signs for all the other values of k are immediately available as well. Yet this very question, which at first seems to be among the easier ones, presents us with unforeseen difficulties, and the method that had guided us this far without impediment utterly refuses any further help.

2.

Before proceeding further, it will be useful to work out some examples of our summation numerically; we should, however, first make a few general observations.

I. In the case where n is a prime number of the form $4m + 1$, let all the quadratic residues of n between 1 and $\frac{1}{2}(n-1)$ inclusive be represented by a', and all the nonresidues between the same limits by b'. All the $n - a'$ are known to be among the a, and all the $n - b'$ among the b. Since all the a', b', $n-a'$, $n-b'$ together evidently make up the entire collection of numbers 1, 2, 3 .... $n-1$, all the a' together with all the $n - a'$ comprise all the a; likewise all the b' together with all the $n - b'$ comprise all the b. Hence

$$\begin{aligned}
\Sigma \cos ak\omega &= \Sigma \cos a'k\omega + \Sigma \cos(n-a')k\omega \\
\Sigma \cos bk\omega &= \Sigma \cos b'k\omega + \Sigma \cos(n-b')k\omega \\
\Sigma \sin ak\omega &= \Sigma \sin a'k\omega + \Sigma \sin(n-a')k\omega \\
\Sigma \sin bk\omega &= \Sigma \sin b'k\omega + \Sigma \sin(n-b')k\omega
\end{aligned}$$

Now, since $\cos(n-a')k\omega = \cos a'k\omega$, $\cos(n-b')k\omega = \cos b'k\omega$, $\sin(n-a')k\omega = -\sin a'k\omega$, $\sin(n-b')k\omega = -\sin b'k\omega$, it follows at once that

$$\begin{aligned}
\Sigma \sin ak\omega &= \Sigma \sin a'k\omega - \Sigma \sin a'k\omega = 0 \\
\Sigma \sin bk\omega &= \Sigma \sin b'k\omega - \Sigma \sin b'k\omega = 0
\end{aligned}$$

The summation of the cosines takes the form

$$\begin{aligned}
\Sigma \cos ak\omega &= 2\Sigma \cos a'k\omega \\
\Sigma \cos bk\omega &= 2\Sigma \cos b'k\omega
\end{aligned}$$

so that we must have

$$\begin{aligned}
1 + 4\Sigma \cos a'k\omega &= \pm\sqrt{n} \\
1 + 4\Sigma \cos b'k\omega &= \mp\sqrt{n} \\
2\Sigma \cos a'k\omega - 2\Sigma \cos b'k\omega &= \pm\sqrt{n}
\end{aligned}$$

II. In the case where n is of the form $4m + 3$, the complement of any residue a to n is a nonresidue, and the complement of any b is a residue. Thus all the $n - a$ coincide with all the b, and all the $n - b$ with all the a. This gives

$$\Sigma \cos ak\omega = \Sigma \cos(n - b)k\omega = \Sigma \cos bk\omega$$

Since all the a and b together make up all the numbers 1, 2, $3 \ldots . n - 1$, and hence $\Sigma \cos ak\omega + \Sigma \cos bk\omega = \cos k\omega + \cos 2k\omega + \cos 3k\omega + \text{etc.} + \cos(n - 1)k\omega = -1$, the summations

$$\Sigma \cos ak\omega = -\tfrac{1}{2}$$
$$\Sigma \cos bk\omega = -\tfrac{1}{2}$$

follow immediately. Likewise,

$$\Sigma \sin ak\omega = \Sigma \sin(n - b)k\omega = -\Sigma \sin bk\omega$$

which shows how the two summations

$$2\Sigma \sin ak\omega = \pm\sqrt{n}$$
$$2\Sigma \sin bk\omega = \mp\sqrt{n}$$

depend on one another.

3.

Here are the numerical calculations for several examples:

I. For $n = 5$ there is one value of a', namely $a' = 1$, and one value of b', namely $b' = 2$. We have

$$\cos \omega = +0{,}3090169944 \qquad \cos 2\omega = -0{,}8090169944$$

and hence $1 + 4 \cos \omega = +\sqrt{5}$, $1 + 4 \cos 2\omega = -\sqrt{5}$.

II. For $n = 13$ there are three values of a', namely 1, 3, 4, and as many values of b', namely 2, 5, 6. We calculate

$\cos \omega$	$= +0{,}8854560257$	$\cos 2\omega$	$= +0{,}5680647467$
$\cos 3\omega$	$= +0{,}1205366803$	$\cos 5\omega$	$= -0{,}7485107482$
$\cos 4\omega$	$= -0{,}3546048870$	$\cos 6\omega$	$= -0{,}9709418174$
Sum	$= +0{,}6513878190$	Sum	$= -1{,}1513878189$

Hence $1 + 4\Sigma \cos a'\omega = +\sqrt{13}$, $1 + 4\Sigma \cos b'\omega = -\sqrt{13}$.

III. For $n = 17$ we have four values of a', namely 1, 2, 4, 8, and as many values of b', namely 3, 5, 6, 7. The cosines are therefore calculated as follows:

$$\begin{array}{rcl} \cos\omega &=& +0{,}9324722294 \\ \cos 2\omega &=& +0{,}7390089172 \\ \cos 4\omega &=& +0{,}0922683595 \\ \cos 8\omega &=& -0{,}9829730997 \\ \hline \text{Sum} &=& +0{,}7807764064 \end{array} \qquad \begin{array}{rcl} \cos 3\omega &=& +0{,}4457383558 \\ \cos 5\omega &=& -0{,}2736629901 \\ \cos 6\omega &=& -0{,}6026346364 \\ \cos 7\omega &=& -0{,}8502171357 \\ \hline \text{Sum} &=& -1{,}2807764065 \end{array}$$

Hence $1 + 4\Sigma \cos a'\omega = +\sqrt{17}, \quad 1 + 4\Sigma \cos b'\omega = -\sqrt{17}$.

IV. For $n = 3$ there is just one value of a, namely $a = 1$, to which corresponds

$$\sin\omega = +0{,}8660254038$$

Hence $2 \sin\omega = +\sqrt{3}$.

V. For $n = 7$ there are three values of a, namely 1, 2, 4. These give the sines

$$\begin{array}{rcl} \sin\omega &=& +0{,}7818314825 \\ \sin 2\omega &=& +0{,}9749279122 \\ \sin 4\omega &=& -0{,}4338837391 \\ \hline \text{Sum} &=& +1{,}3228756556, \end{array}$$ and hence $2\Sigma \sin a\omega = +\sqrt{7}$.

VI. For $n = 11$ the values of a are 1, 3, 4, 5, 9, to which correspond the sines

$$\begin{array}{rcl} \sin\omega &=& +0{,}5406408175 \\ \sin 3\omega &=& +0{,}9898214419 \\ \sin 4\omega &=& +0{,}7557495744 \\ \sin 5\omega &=& +0{,}2817325568 \\ \sin 9\omega &=& -0{,}9096319954 \\ \hline \text{Sum} &=& +1{,}6583123952, \end{array}$$ and hence $2\Sigma \sin a\omega = +\sqrt{11}$.

VII. For $n = 19$ the values of a are 1, 4, 5, 6, 7, 9, 11, 16, 17, to which correspond the sines

$$\begin{aligned}
\sin\omega &= +0{,}3246994692\\
\sin 4\omega &= +0{,}9694002659\\
\sin 5\omega &= +0{,}9965844930\\
\sin 6\omega &= +0{,}9157733267\\
\sin 7\omega &= +0{,}7357239107\\
\sin 9\omega &= +0{,}1645945903\\
\sin 11\omega &= -0{,}4759473930\\
\sin 16\omega &= -0{,}8371664783\\
\sin 17\omega &= -0{,}6142127127\\
\text{Sum} &= +2{,}1794494718,
\end{aligned}$$

and hence $2\Sigma \sin a\omega = +\sqrt{19}$.

4.

In all these examples the radical takes the positive sign. This is also easily confirmed for the larger values $n = 23$, $n = 29$, etc., and already makes it highly probable that the same holds in general. But a proof of this phenomenon cannot be drawn from the principles set out in the place cited, and it must with full justification be regarded as a matter for deeper investigation. The purpose of this paper is therefore to present a rigorous proof of this most elegant theorem, attempted in vain in various ways over several years and finally achieved through unusual and quite subtle considerations. At the same time we shall bring the theorem to a far greater degree of generality, preserving, or rather increasing, its elegance. Finally, as a crowning addition, we shall show the remarkable and very close connection between this summation and another arithmetical theorem of the greatest importance. We hope that these investigations will themselves be welcome to geometers, and that the methods by which we have accomplished all this, and which may also be useful on other occasions, will be considered worthy of their attention.

5.

Our proof comes from considering a special kind of series whose terms depend on expressions such as

$$\frac{(1-x^m)\,(1-x^{m-1})\,(1-x^{m-2})\ldots.(1-x^{m-\mu+1})}{(1-x)\,(1-xx)\,(1-x^3)\ldots..(1-x^\mu)}$$

For brevity we shall denote such a fraction by (m, μ), and first make a few general observations about functions of this kind.

I. Whenever m is a positive integer less than μ, the function (m, μ) evidently vanishes, since the numerator contains the factor $1 - x^0$. For $m = \mu$, the factors in the numerator are identical, in reverse order, to those in the denominator, so that $(\mu, \mu) = 1$. Finally, for the case where m is a positive integer greater than μ, we have the formulas

$$(\mu+1,\mu) = \frac{1-x^{\mu+1}}{1-x} = (\mu+1,1)$$
$$(\mu+2,\mu) = \frac{(1-x^{\mu+2})\,(1-x^{\mu+1})}{(1-x)\qquad(1-xx)} = (\mu+2,2)$$
$$(\mu+3,\mu) = \frac{(1-x^{\mu+3})(1-x^{\mu+2})(1-x^{\mu+1})}{(1-x)\qquad(1-xx)\qquad(1-x^3)} = (\mu+3,3) \text{ etc.}$$

or, in general,

$$(m,\mu) = (m, m-\mu)$$

II. Further, it is easily confirmed that in general

$$(m,\mu+1) = (m-1,\mu+1) + x^{m-\mu-1}(m-1,\mu)$$

and since we likewise have

$$(m-1,\mu+1) = (m-2,\mu+1) + x^{m-\mu-2}(m-2,\mu)$$
$$(m-2,\mu+1) = (m-3,\mu+1) + x^{m-\mu-3}(m-3,\mu)$$
$$(m-3,\mu+1) = (m-4,\mu+1) + x^{m-\mu-4}(m-4,\mu), \text{ etc.}$$

and can continue this series as far as

$$\begin{aligned}(\mu+2,\mu+1) &= (\mu+1,\mu+1) + x(\mu+1,\mu)\\ &= (\mu,\mu) + x(\mu+1,\mu)\end{aligned}$$

provided that m is a positive integer greater than $\mu+1$, we obtain

$$\begin{aligned}(m,\mu+1) = (\mu,\mu) &+ x(\mu+1,\mu) + xx(\mu+2,\mu)\\ &+ x^3(\mu+3,\mu) + \text{etc.}\\ &+ x^{m-\mu-1}(m-1,\mu)\end{aligned}$$

This shows that if, for some fixed value of μ, every function (m, μ) is a polynomial whenever m is a positive integer, then every function $(m, \mu + 1)$ must also be a polynomial. Since that supposition holds for $\mu = 1$, it therefore also holds for $\mu = 2$, and hence for $\mu = 3$, etc. That is, in general, for any positive integral value of m, (m, μ) is a polynomial, or equivalently the product

$$(1 - x^m)(1 - x^{m-1})(1 - x^{m-2}) \dots (1 - x^{m-\mu+1})$$

is divisible by

$$(1 - x)(1 - x^2)(1 - x^3) \dots (1 - x^\mu)$$

6.

We shall now consider two series, both of which can lead to our goal. The first series is

$$1 - \frac{1-x^m}{1-x} + \frac{(1-x^m)\,(1-x^{m-1})}{(1-x)\,(1-xx)} - \frac{(1-x^m)(1-x^{m-1})(1-x^{m-2})}{(1-x)\quad(1-xx)\quad(1-x^3)} + \text{etc.}$$

or

$$1 - (m,1) + (m,2) - (m,3) + (m,4) - \text{etc.}$$

which for brevity we shall denote by $f(x,m)$. First, it is immediately evident that whenever m is a positive integer this series *terminates* after its term in position $m+1$ (with $= \pm 1$), and that in this case its sum must therefore be a finite polynomial in x. Further, Article 5, II, shows that in general, for any value of m,

$$\begin{aligned} 1 &= 1 \\ -(m,1) &= -(m-1,1) - x^{m-1} \\ +(m,2) &= +(m-1,2) + x^{m-2}(m-1,1) \\ -(m,3) &= -(m-1,3) - x^{m-3}(m-1,2), \text{ etc.} \end{aligned}$$

and hence

$$\begin{aligned} f(x,m) = 1 - x^{m-1} &- (1-x^{m-2})(m-1,1) \\ &+ (1-x^{m-3})(m-1,2) \\ &- (1-x^{m-4})(m-1,3) + \text{etc.} \end{aligned}$$

But evidently

$$\begin{aligned} (1-x^{m-2})(m-1,1) &= (1-x^{m-1})(m-2,1) \\ (1-x^{m-3})(m-1,2) &= (1-x^{m-1})(m-2,2) \\ (1-x^{m-4})(m-1,3) &= (1-x^{m-1})(m-2,3), \text{ etc.} \end{aligned}$$

from which we deduce the equation

$$f(x,m) = (1-x^{m-1})f(x,m-2) \ldots\ldots\ldots [1]$$

7.

Since for $m = 0$ we have $f(x, m) = 1$, the formula just found gives

$$\begin{aligned} f(x,2) &= 1 - x \\ f(x,4) &= (1-x)(1-x^3) \\ f(x,6) &= (1-x)(1-x^3)(1-x^5) \\ f(x,8) &= (1-x)(1-x^3)(1-x^5)(1-x^7), \text{ etc.} \end{aligned}$$

or, in general, for any even value of m,

$$f(x,m) = (1-x)(1-x^3)(1-x^5)\ldots.(1-x^{m-1}) \ldots [2].$$

Conversely, since for $m = 1$ we have $f(x, m) = 0$, we also have

$$\begin{aligned} f(x,3) &= 0 \\ f(x,5) &= 0 \\ f(x,7) &= 0 \text{ etc.} \end{aligned}$$

or, in general, for any odd value of m,

$$f(x,m) = 0$$

The latter summation could, moreover, already have been derived from the fact that in the series

$$1 - (m,1) + (m,2) - (m,3) + \text{etc.} + (m,m-1) - (m,m)$$

the last term cancels the first, the next to last cancels the second, etc.

8.

For our purpose the case where m is a positive odd integer is sufficient. But the unusual nature of the matter makes it worthwhile to add a few remarks about the cases where m is fractional or negative. Our series then evidently no longer terminates, but continues to infinity; it is also easily seen that it becomes divergent whenever x is given a value less than 1, so that its summation must be restricted to values of x greater than 1.

By formula [1] of Article 6 we have

$$f(x,-2) = \frac{1}{1-\frac{1}{x}}$$

$$f(x,-4) = \frac{1}{1-\frac{1}{x}} \cdot \frac{1}{1-\frac{1}{x^3}}$$

$$f(x,-6) = \frac{1}{1-\frac{1}{x}} \cdot \frac{1}{1-\frac{1}{x^3}} \cdot \frac{1}{1-\frac{1}{x^5}}, \text{ etc.}$$

so that the value of the function $f(x,m)$ can also be given in finite terms for a negative even integral value of m. For the remaining values of m, however, we shall convert the function $f(x,m)$ into an *infinite product* as follows.

As m tends to a negative *infinite* value, the function $f(x,m)$ becomes

$$1 + \frac{1}{x-1} + \frac{1}{x-1} \cdot \frac{1}{xx-1} + \frac{1}{x-1} \cdot \frac{1}{xx-1} \cdot \frac{1}{x^3-1} + \text{etc.}$$

This series is therefore equal to the infinite product

$$\frac{1}{1-\frac{1}{x}} \cdot \frac{1}{1-\frac{1}{x^3}} \cdot \frac{1}{1-\frac{1}{x^5}} \cdot \frac{1}{1-\frac{1}{x^7}} \text{ etc. to infinity.}$$

Further, since in general

$$f(x,m) = f(x,m-2\lambda) \,.\, (1-x^{m-1})(1-x^{m-3}) \\ (1-x^{m-5}) \,..\, (1-x^{m-2\lambda+1})$$

we have

$$f(x,m) = f(x,-\infty) \,.\, (1-x^{m-1})(1-x^{m-3})(1-x^{m-5}) \\ \text{etc. to infinity.} \\ = \frac{1-x^{m-1}}{1-x^{-1}} \cdot \frac{1-x^{m-3}}{1-x^{-3}} \cdot \frac{1-x^{m-5}}{1-x^{-5}} \cdot \frac{1-x^{m-7}}{1-x^{-7}} \\ \text{etc. to infinity.}$$

whose factors evidently converge ever more closely to unity as they continue.

The case $m = -1$ deserves particular attention. Here

$$f(x,-1) = 1 + x^{-1} + x^{-3} + x^{-6} + x^{-10} + \text{etc.}$$

This series is therefore equal to the infinite product

$$\frac{1-x^{-2}}{1-x^{-1}} \cdot \frac{1-x^{-4}}{1-x^{-3}} \cdot \frac{1-x^{-6}}{1-x^{-5}} \text{ etc.}$$

or, writing x for x^{-1}, we obtain

$$1+x+x^3+x^6+\text{etc.} = \frac{1-xx}{1-x} \cdot \frac{1-x^4}{1-x^3} \cdot \frac{1-x^6}{1-x^5} \cdot \frac{1-x^8}{1-x^7} \text{ etc.}$$

This equality between two rather intricate expressions, to which we shall return on another occasion, is certainly very remarkable.

9.

Secondly, we shall consider the series

$$1 + x^{\frac{1}{2}}\frac{1-x^m}{1-x} + x\frac{(1-x^m)(1-x^{m-1})}{(1-x)(1-xx)} + x^{\frac{3}{2}}\frac{(1-x^m)(1-x^{m-1})(1-x^{m-2})}{(1-x)(1-xx)(1-x^3)} + \text{etc.}$$

or

$$1 + x^{\frac{1}{2}}(m,1) + x(m,2) + x^{\frac{3}{2}}(m,3) + xx(m,4) + \text{etc.}$$

which we shall denote by $F(x,m)$. We restrict this investigation to the case where m is a positive integer, so that this series too always ends at its term in position $m+1$ (with $= x^{\frac{1}{2}m}(m,m)$). Since

$$(m,m) = 1, \quad (m,m-1) = (m,1), \quad (m,m-2) = (m,2), \text{ etc.}$$

the series can also be written as follows:

$$F(x,m) = x^{\frac{1}{2}m} + x^{\frac{1}{2}(m-1)}(m,1) + x^{\frac{1}{2}(m-2)}(m,2) + x^{\frac{1}{2}(m-3)}(m,3) + \text{etc.}$$

This gives

$$\begin{aligned}(1+x^{\frac{1}{2}m+\frac{1}{2}})F(x,m) = 1 &+ x^{\frac{1}{2}}(m,1) + x(m,2) \\ &+ x^{\frac{3}{2}}(m,3) + \text{etc.} \\ &+ x^{\frac{1}{2}} \,.\, x^m + x \,.\, x^{m-1}(m,1) \\ &+ x^{\frac{3}{2}} \,.\, x^{m-2}(m,2) + \text{etc.}\end{aligned}$$

Since we have (Article 5, II)

$$\begin{aligned}(m,1)+x^m &= (m+1,1)\\ (m,2)+x^{m-1}(m,1) &= (m+1,2)\\ (m,3)+x^{m-2}(m,2) &= (m+1,3),\ \text{etc.}\end{aligned}$$

we obtain

$$(1+x^{\frac{1}{2}m+\frac{1}{2}})F(x,m) = F(x,m+1)\ \ldots\ldots\ [3]$$

But $F(x,0)=1$, so that

$$\begin{aligned}F(x,1) &= 1+x^{\frac{1}{2}}\\ F(x,2) &= (1+x^{\frac{1}{2}})(1+x)\\ F(x,3) &= (1+x^{\frac{1}{2}})(1+x)(1+x^{\frac{3}{2}}),\ \text{etc.}\end{aligned}$$

or, in general,

$$\begin{aligned}F(x,m) = (1+x^{\frac{1}{2}})&(1+x)(1+x^{\frac{3}{2}})\\ &\ldots.(1+x^{\frac{1}{2}m})\ \ldots.\quad [4]\end{aligned}$$

10.

With these preliminary investigations in place, let us now come closer to our aim. For a prime value of n, the squares 1, 4, $9\ldots.(\frac{1}{2}(n-1))^2$ are all incongruent to one another modulo n; their least residues for this modulus must therefore be identical to the numbers a, and hence

$$\begin{aligned}\Sigma\cos ak\omega &= \cos k\omega+\cos 4k\omega+\cos 9k\omega\\ &\quad+\text{etc.}+\cos(\tfrac{1}{2}(n-1))^2k\omega\\ \Sigma\sin ak\omega &= \sin k\omega+\sin 4k\omega+\sin 9k\omega\\ &\quad+\text{etc.}+\sin(\tfrac{1}{2}(n-1))^2k\omega\end{aligned}$$

Likewise, since the same squares 1, 4, $9\ldots.(\frac{1}{2}(n-1))^2$, in reverse order, are congruent to $(\frac{1}{2}(n+1))^2$, $(\frac{1}{2}(n+3))^2$, $(\frac{1}{2}(n+5))^2\ldots.(n-1)^2$, we also have

$$\begin{aligned}\Sigma\cos ak\omega &= \cos(\tfrac{1}{2}(n+1))^2k\omega\\ &\quad+\cos(\tfrac{1}{2}(n+3))^2k\omega+\text{etc.}+\cos(n-1)^2k\omega\\ \Sigma\sin ak\omega &= \sin(\tfrac{1}{2}(n+1))^2k\omega\\ &\quad+\sin(\tfrac{1}{2}(n+3))^2k\omega+\text{etc.}+\sin(n-1)^2k\omega\end{aligned}$$

Thus, putting

$$\begin{aligned} T &= 1 + \cos k\omega + \cos 4k\omega + \cos 9k\omega \\ &\quad + \text{etc.} + \cos(n-1)^2 k\omega \\ U &= \quad \sin k\omega + \sin 4k\omega + \sin 9k\omega \\ &\quad + \text{etc.} + \sin(n-1)^2 k\omega \end{aligned}$$

we obtain

$$\begin{aligned} 1 + 2\Sigma \cos ak\omega &= T \\ 2\Sigma \sin ak\omega &= U \end{aligned}$$

This shows that the summations proposed in Article 1 depend on summing the series T and U. We shall therefore leave the former aside and direct our investigation to these, carrying it out generally enough to include both prime values of n and any composite values. We shall, however, suppose that the number k is relatively prime to n: the case where k and n have a common divisor can be reduced to this one without difficulty.

11.

Let us denote the imaginary quantity $\sqrt{-1}$ by i and put

$$\cos k\omega + i \sin k\omega = r$$

Then $r^n = 1$, or r is a root of the equation $r^n - 1 = 0$. It is easy to see that all the numbers k, $2k$, $3k \ldots . (n-1)k$ are not divisible by n and are incongruent to one another modulo n. Thus the powers of r

$$1,\ r,\ rr,\ r^3 \ldots . r^{n-1}$$

are all unequal, while each also satisfies the equation $x^n - 1 = 0$. For this reason these powers represent *all* the roots of the equation $x^n - 1 = 0$.

These conclusions would not hold if k had a common divisor with n. For if ν were such a common divisor, $k \,.\, \dfrac{n}{\nu}$ would be divisible by n, and hence a power lower than r^n, namely $r^{\frac{n}{\nu}}$, would equal unity. In this case, therefore, the powers of r represent at most $\dfrac{n}{\nu}$ roots of the equation $x^n - 1 = 0$, and indeed give that many distinct roots if ν is the *greatest* common divisor of the numbers k, n. In our case, where k and n are supposed relatively prime, r can conveniently be called a *proper root* of the equation $x^n - 1 = 0$. In the other case, where k and n have a (greatest) common divisor ν, r would be called an *improper root* of that equation. Then, however, the same r would evidently be a proper root of the equation $x^{\frac{n}{\nu}} - 1 = 0$. The simplest improper root is unity, and when n is a prime number there are no other improper roots at all.

12.

Now, if we put

$$W = 1 + r + r^4 + r^9 + \text{etc.} + \text{etc.} + r^{(n-1)^2}$$

we evidently have $W = T + iU$, so that T is the real part of W, and U is obtained from the imaginary part of W by removing the factor i. The whole question is therefore reduced to finding the sum W. For this purpose we can use either the series considered in Article 6 or the one we showed how to sum in Article 9; the former, however, is less suitable when n is even. Nevertheless, we hope that readers will welcome our treating the case where n is odd by two methods.

First, then, suppose that n is an odd number, that r denotes any proper root of the equation $x^n - 1 = 0$, and that in the function $f(x, m)$ we put $x = r$ and $m = n - 1$. We evidently obtain

$$\begin{aligned}
\frac{1 - x^m}{1 - x} &= \frac{1 - r^{-1}}{1 - r} = -r^{-1} \\
\frac{1 - x^{m-1}}{1 - xx} &= \frac{1 - r^{-2}}{1 - rr} = -r^{-2} \\
\frac{1 - x^{m-2}}{1 - x^3} &= \frac{1 - r^{-3}}{1 - r^3} = -r^{-3}, \text{ etc.}
\end{aligned}$$

as far as

$$\frac{1 - x}{1 - x^m} = \frac{1 - r^{-m}}{1 - r^m} = -r^{-m}$$

(It is worth pointing out that these equations hold only insofar as r is supposed to be a proper root: if r were an improper root, the numerator and denominator of some of these fractions would vanish together, and the fractions would therefore become indeterminate).

From this we deduce the following equation

$$\begin{aligned}
f(r, n-1) &= 1 + r^{-1} + r^{-3} + r^{-6} + \text{etc.} + r^{-\frac{1}{2}(n-1)n} \\
&= (1 - r)(1 - r^3)(1 - r^5) \ldots . (1 - r^{n-2})
\end{aligned}$$

The same equation still holds if we replace r by r^λ, where λ denotes any integer relatively prime to n: for then r^λ too is a proper root of the equation $x^n - 1 = 0$. Let us therefore replace r by r^{n-2}, or equivalently r^{-2}. We obtain

$$\begin{aligned}
&1 + r^2 + r^6 + r^{12} + \text{etc.} + r^{(n-1)n} \\
&\quad = (1 - r^{-2})(1 - r^{-6})(1 - r^{-10}) \ldots (1 - r^{-2(n-2)})
\end{aligned}$$

Let us multiply both sides of this equation by

$$r \,.\, r^3 \,.\, r^5 \,.\,.\,.\,. \, r^{(n-2)} = r^{\frac{1}{4}(n-1)^2}$$

and, since

$$\begin{aligned}
r^{2+\frac{1}{4}(n-1)^2} &= r^{\frac{1}{4}(n-3)^2}, \\
r^{(n-1)n+\frac{1}{4}n-1)^2} &= r^{\frac{1}{4}(n+1)^2} \\
r^{6+\frac{1}{4}(n-1)^2} &= r^{\frac{1}{4}(n-5)^2}, \\
r^{(n-2)(n-1)+\frac{1}{4}(n-1)^2} &= r^{\frac{1}{4}(n+3)^2} \\
r^{12+\frac{1}{4}(n-1)^2} &= r^{\frac{1}{4}(n-7)^2}, \\
r^{(n-3)(n-2)+\frac{1}{4}(n-1)^2} &= r^{\frac{1}{4}(n+5)^2}, \text{ etc.}
\end{aligned}$$

the following equation results:

$$\begin{aligned}
&r^{\frac{1}{4}(n-1)^2} + r^{\frac{1}{4}(n-3)^2} + r^{\frac{1}{4}(n-5)^2} + \text{etc.} + r + 1 \\
&+ r^{\frac{1}{4}(n+1)^2} + r^{\frac{1}{4}(n+3)^2} + r^{\frac{1}{4}(n+5)^2} \\
&\qquad + \text{etc.} + r^{\frac{1}{4}(2n-2)^2} \\
&\qquad = (r - r^{-1})(r^3 - r^{-3})(r^5 - r^{-5}) \,.\,.\,.\,. \, (r^{n-2} - r^{-n+2})
\end{aligned}$$

or, arranging the terms on the left side differently,

$$\begin{aligned}
&1 + r + r^4 + \text{etc.} + r^{(n-1)^2} \\
&\qquad = (r - r^{-1})(r^3 - r^{-3}) \,.\,.\, (r^{n-2} - r^{-n+2}) \quad .. \quad [5]
\end{aligned}$$

13.

The factors on the right side of equation [5] can also be written as follows:

$$r - r^{-1} = -(r^{n-1} - r^{-n+1})$$
$$r^3 - r^{-3} = -(r^{n-3} - r^{-n+3})$$
$$r^5 - r^{-5} = -(r^{n-5} - r^{-n+5}), \text{ etc.}$$

up to

$$r^{n-2} - r^{-n+2} = -(r^2 - r^{-2})$$

The equation then takes this form:

$$W = (-1)^{\frac{1}{2}(n-1)}(r^2 - r^{-2})(r^4 - r^{-4})(r^6 - r^{-6}) \\ \ldots . (r^{n-1} - r^{-n+1})$$

Multiplying this equation by [5] in its original form gives

$$W^2 = (-1)^{\frac{1}{2}(n-1)}(r - r^{-1})(r^2 - r^{-2})(r^3 - r^{-3}) \\ \ldots . (r^{n-1} - r^{-n+1})$$

where $(-1)^{\frac{1}{2}(n-1)}$ is either $= +1$ or $= -1$, according as n is of the form $4\mu + 1$ or of the form $4\mu + 3$. Hence

$$W^2 = \pm r^{\frac{1}{2}n(n-1)}(1 - r^{-2})(1 - r^{-4})(1 - r^{-6}) \\ \ldots . (1 - r^{-2(n-1)})$$

But it is easy to see that r^{-2}, r^{-4}, $r^{-6} \ldots . r^{-2n+2}$ give all the roots of the equation $x^n - 1 = 0$ except the root $x = 1$. The following identity must therefore hold for an indeterminate:

$$(x - r^{-2})(x - r^{-4})(x - r^{-6}) \ldots . (x - r^{-2n+2}) \\ = x^{n-1} + x^{n-2} + x^{n-3} + \text{etc.} + x + 1$$

Thus, putting $x = 1$, we obtain

$$(1 - r^{-2})(1 - r^{-4})(1 - r^{-6}) \ldots . (1 - r^{-2n+2}) = n$$

and since evidently $r^{\frac{1}{2}n(n-1)} = 1$, our equation becomes

$$W^2 = \pm n \ldots . \ldots . \ldots . [6]$$

Thus, in the case where n is of the form $4\mu + 1$, we have

$$W = \pm\sqrt{n}, \quad \text{and hence} \quad T = \pm\sqrt{n}, \quad U = 0$$

In the other case, where n is of the form $4\mu + 3$, we instead have

$$W = \pm i\sqrt{n}, \quad \text{and so} \quad T = 0, \quad U = \pm\sqrt{n}$$

14.

The method of the preceding article gives only the absolute value of the sums T, U, and leaves it uncertain whether T in the first case and U in the second should be set $= +\sqrt{n}$ or $= -\sqrt{n}$. But, at least for the case where $k = 1$, this can be decided from equation [5] as follows. Since, for $k = 1$,

$$\begin{aligned} r - r^{-1} &= 2i \sin \omega \\ r^3 - r^{-3} &= 2i \sin 3\omega \\ r^5 - r^{-5} &= 2i \sin 5\omega \text{ etc.} \end{aligned}$$

that equation becomes

$$W = (2i)^{\frac{1}{2}(n-1)} \sin\omega \sin 3\omega \sin 5\omega \ldots \sin(n-2)\omega$$

Now, in the case where n is of the form $4\mu+1$, the sequence of odd numbers

$$1,\ 3,\ 5,\ 7 \ldots . \tfrac{1}{2}(n-3), \quad \tfrac{1}{2}(n+1) \ldots . (n-2)$$

contains $\frac{1}{4}(n-1)$ numbers smaller than $\frac{1}{2}n$, to which positive sines evidently correspond. The remaining $\frac{1}{4}(n-1)$, on the other hand, are greater than $\frac{1}{2}n$, and negative sines correspond to them. The product of all the sines must therefore equal a positive quantity multiplied by $(-1)^{\frac{1}{4}(n-1)}$. Thus W will equal a positive real quantity multiplied by i^{n-1}, or by 1, since $i^4 = 1$ and $n-1$ is divisible by 4: that is, W will be real and positive. Hence we must have

$$W = +\sqrt{n}, \quad T = +\sqrt{n}$$

In the other case, where n is of the form $4\mu + 3$, in the sequence of odd numbers

$$1,\ 3,\ 5,\ 7 \ldots . \tfrac{1}{2}(n-1), \quad \tfrac{1}{2}(n+3) \ldots . (n-2)$$

the first $\frac{1}{4}(n+1)$ will be smaller than $\frac{1}{2}n$, but the remaining $\frac{1}{4}(n-3)$ will be greater. Thus, among the sines of the arcs ω, 3ω, $5\omega \ldots . (n-2)\omega$, there will be $\frac{1}{4}(n-3)$ negative ones, and so W will be the product of $i^{\frac{1}{2}(n-1)}$, a positive real quantity and $(-1)^{\frac{1}{4}(n-3)}$. The third factor is $= i^{\frac{1}{2}(n-3)}$, which, multiplied by the first, gives $i^{n-2} = i$, since $i^{n-3} = 1$. We must therefore have

$$W = +i\sqrt{n}, \quad \text{and} \quad U = +\sqrt{n}$$

15.

We shall now show how the same conclusions can be derived from the series considered in Article 9. In equation [4], replace $x^{\frac{1}{2}}$ by $-y^{-1}$; we then obtain

$$1 - y^{-1}\frac{1-y^{-2m}}{1-y^{-2}} + y^{-2}\frac{(1-y^{-2m})(1-y^{-2m+2})}{(1-y^{-2})(1-y^{-4})} - y^{-3}\frac{(1-y^{-2m})(1-y^{-2m+2})(1-y^{-2m+4})}{(1-y^{-2})(1-y^{-4})(1-y^{-6})} + \text{etc.}$$

up to the term in position $m+1$

$$= (1-y^{-1})(1+y^{-2})(1-y^{-3})(1+y^{-4}) \ldots (1 \pm y^{-m}) \ldots . \text{ [7]}$$

If we take for y a proper root of the equation $y^n - 1 = 0$, say r, and at the same time put $m = n-1$, we obtain

$$\frac{1-y^{-2m}}{1-y^{-2}} = \frac{1-r^2}{1-r^{-2}} = -r^2$$
$$\frac{1-y^{-2m+2}}{1-y^{-4}} = \frac{1-r^4}{1-r^{-4}} = -r^4$$
$$\frac{1-y^{-2m+4}}{1-y^{-6}} = \frac{1-r^6}{1-r^{-6}} = -r^6$$

up to

$$\frac{1-y^{-2}}{1-y^{-2m}} = \frac{1-r^{2n-2}}{1-r^{-2n+2}} = -r^{2n-2}$$

Here we should note that none of the denominators $1-r^{-2}$, $1-r^{-4}$, etc. becomes $= 0$. Equation [7] therefore takes this form:

$$1 + r + r^4 + r^9 + \text{etc.} + r^{(n-1)^2} = (1-r^{-1})(1+r^{-2})(1-r^{-3}) \ldots . (1+r^{-n+1})$$

Multiplying, on the right side of this equation, the first term by the last, the second by the next to last, etc., we obtain

$$(1-r^{-1})(1+r^{-n+1}) = r - r^{-1}$$
$$(1+r^{-2})(1-r^{-n+2}) = r^{n-2} - r^{-n+2}$$
$$(1-r^{-3})(1+r^{-n+3}) = r^3 - r^{-3}$$
$$(1+r^{-4})(1-r^{-n+4}) = r^{n-4} - r^{-n+4} \text{ etc.}$$

It is easy to see that these partial products combine to form the product

$$(r - r^{-1})(r^3 - r^{-3})(r^5 - r^{-5}) \\ \ldots . (r^{n-4} - r^{-n+4})(r^{n-2} - r^{-n+2})$$

which will therefore be

$$= 1 + r + r^4 + r^9 + \text{etc.} + r^{(n-1)^2} = W$$

This equation is identical with equation [5], derived in Article 12 from the first series; the remaining arguments are then established in the same way as in Articles 13 and 14.

16.

We turn to the other case, where n is an even number. First let n be of the form $4\mu + 2$, or twice an odd number. It is clear that the numbers $\frac{1}{4}nn$, $(\frac{1}{2}n + 1)^2 - 1$, $(\frac{1}{2}n + 2)^2 - 4$, etc., or in general $(\frac{1}{2}n + \lambda)^2 - \lambda\lambda$, give odd quotients when divided by $\frac{1}{2}n$, and are therefore congruent, modulo n, to $\frac{1}{2}n$. It follows that, if r is a proper root of the equation $x^n - 1 = 0$, and hence $r^{\frac{1}{2}n} = -1$, we have

$$\begin{aligned} r^{(\frac{1}{2}n)^2} &= -1 \\ r^{(\frac{1}{2}n+1)^2} &= -r \\ r^{(\frac{1}{2}n+2)^2} &= -r^4 \\ r^{(\frac{1}{2}n+2)^2} &= -r^9 \text{ etc.} \end{aligned}$$

Thus, in the series

$$1 + r + r^4 + r^9 + \text{etc.} + r^{(n-1)^2}$$

the term $r^{(\frac{1}{2}n)^2}$ cancels the first, the next cancels the second, etc., so that

$$W = 0, \quad T = 0, \quad U = 0$$

17.

The case remains where n is of the form 4μ, or divisible by four. Here, in general, $(\frac{1}{2}n+\lambda)^2 - \lambda\lambda$ is divisible by n, and therefore

$$r^{(\frac{1}{2}n+\lambda)^2} = r^{\lambda\lambda}$$

Thus, in the series

$$1 + r + r^4 + r^9 + \text{etc.} + r^{(n-1)^2}$$

the term $r^{(\frac{1}{2}n)^2}$ equals the first, the next equals the second, etc., so that

$$W = 2(1 + r + r^4 + r^9 + \text{etc.} + r^{(\frac{1}{2}n-1)^2})$$

Now suppose that in equation [7] of Article 15 we put $m = \frac{1}{2}n - 1$, and take for y a proper root of the equation $y^n - 1 = 0$, say r. Then, just as in Article 15, the equation takes the following form:

$$1 + r + r^4 + \text{etc.} + r^{(\frac{1}{2}n-1)^2} = (1 - r^{-1})(1 + r^{-2})(1 - r^{-3}) \ldots . (1 - r^{-\frac{1}{2}n+1})$$

or

$$W = 2(1 - r^{-1})(1 + r^{-2})(1 - r^{-3})(1 + r^{-4}) \ldots (1 - r^{-\frac{1}{2}n+1}) \quad .. \quad [8]$$

Further, since $r^{\frac{1}{2}n} = -1$, and therefore

$$1 + r^{-2} = -r^{\frac{1}{2}n-2}(1 - r^{-\frac{1}{2}n+2})$$
$$1 + r^{-4} = -r^{\frac{1}{2}n-4}(1 - r^{-\frac{1}{2}n+4})$$
$$1 + r^{-6} = -r^{\frac{1}{2}n-6}(1 - r^{-\frac{1}{2}n+6}) \text{ etc.}$$

and the product of the factors $-r^{\frac{1}{2}n-2}$, $-r^{\frac{1}{2}n-4}$, $-r^{\frac{1}{2}n-6}$, etc., up to $-r^2$ is $= (-1)^{\frac{1}{4}n-1} r^{\frac{1}{16}nn-\frac{1}{4}n}$, the preceding equation can also be written as

$$W = 2(-1)^{\frac{1}{4}n-1} r^{\frac{1}{16}nn-\frac{1}{4}n}(1 - r^{-1})(1 - r^{-2}) (1 - r^{-3})(1 - r^{-4}) \ldots . (1 - r^{-\frac{1}{2}n+1})$$

Since we have

$$1 - r^{-1} = -r^{-1}(1 - r^{-n+1})$$
$$1 - r^{-2} = -r^{-2}(1 - r^{-n+2})$$
$$1 - r^{-3} = -r^{-3}(1 - r^{-n+3}) \text{ etc.}$$

we obtain

$$(1 - r^{-1})(1 - r^{-2})(1 - r^{-3}) \dots (1 - r^{-\frac{1}{2}n+1})$$
$$= (-1)^{\frac{1}{2}n-1} r^{-\frac{1}{8}nn+\frac{1}{4}n}(1 - r^{-\frac{1}{2}n-1})$$
$$(1 - r^{-\frac{1}{2}n-2})(1 - r^{-\frac{1}{2}n-3}) \dots (1 - r^{-n+1})$$

and hence

$$W = 2(-1)^{\frac{3}{4}n-2} r^{-\frac{1}{16}nn}(1 - r^{-\frac{1}{2}n-1})$$
$$(1 - r^{-\frac{1}{2}n-2})(1 - r^{-\frac{1}{2}n-3}) \dots . (1 - r^{-n+1})$$

Multiplying this value of W by the one found earlier, and introducing the factor $1 - r^{-\frac{1}{2}n}$ on both sides, gives

$$(1 - r^{-\frac{1}{2}n})W^2 = 4(-1)^{n-3} r^{-\frac{1}{4}n}(1 - r^{-1}.)$$
$$(1 - r^{-2})(1 - r^{-3}) \dots . (1 - r^{-n+1})$$

But we have

$$1 - r^{-\frac{1}{2}n} = 2$$
$$(-1)^{n-3} = -1$$
$$r^{-\frac{1}{4}n} = -r^{\frac{1}{4}n}$$

$$(1 - r^{-1})(1 - r^{-2})(1 - r^{-3}) \dots . (1 - r^{-n+1}) = n$$

From this we finally conclude

$$W^2 = 2r^{\frac{1}{4}n} n \text{ . . . . . [9]}$$

Now it is easy to see that $r^{\frac{1}{4}n}$ is either $= +i$ or $= -i$, according as k is of the form $4\mu + 1$ or of the form $4\mu + 3$. And since

$$2i = (1 + i)^2, \quad -2i = (1 - i)^2$$

in the case where k is of the form $4\mu + 1$, we have

$$W = \pm(1 + i)\sqrt{n}, \quad \text{and so} \quad T = U = \pm\sqrt{n}$$

but in the other case, where k is of the form $4\mu + 3$,

$$W = \pm(1-i)\sqrt{n}, \quad \text{and so} \quad T = -U = \pm\sqrt{n}$$

18.

The method of the preceding article supplied the absolute values of the functions T, U, and specified the conditions under which they must be given equal or opposite signs; but it does not yet determine the signs themselves. For the case where $k = 1$, we shall supply what is missing as follows.

Let us put $\rho = \cos\frac{1}{2}\omega + i\sin\frac{1}{2}\omega$, so that $r = \rho\rho$. Since $\rho^n = -1$, it is clear that equation [8] can be written as

$$W = 2(1+\rho^{n-2})(1+\rho^{-4})(1+\rho^{n-6})(1+\rho^{-8})$$
$$\ldots(1+\rho^{-n+4})(1+\rho^2)$$

or, with the factors arranged in a different order,

$$W = 2(1+\rho^2)(1+\rho^{-4})(1+\rho^6)(1+\rho^{-8})$$
$$\ldots(1+\rho^{-n+4})(1+\rho^{n-2})$$

Now we have

$$1+\rho^2 = 2\rho\cos\frac{1}{2}\omega$$
$$1+\rho^{-4} = 2\rho^{-2}\cos\omega$$
$$1+\rho^{-6} = 2\rho^3\cos\frac{3}{2}\omega$$
$$1+\rho^{-8} = 2\rho^{-4}\cos 2\omega, \text{ etc.}$$

up to

$$1+\rho^{-n+4} = 2\rho^{-\frac{1}{2}n+2}, \cos(\frac{1}{4}n-1)\omega$$
$$1+\rho^{n-2} = 2\rho^{\frac{1}{2}n-1}\cos(\frac{1}{4}n-\frac{1}{2})\omega$$

Thus we have

$$W = 2^{\frac{1}{2}n}\rho^{\frac{1}{4}n}\cos\frac{1}{2}\omega\cos\omega\,\frac{3}{2}\omega$$
$$\ldots\cos(\frac{1}{4}n-\frac{1}{2})\omega$$

The cosines entering this product are evidently all positive, while the factor $\rho^{\frac{1}{4}n}$ is $= \cos 45^0 + i \sin 45^0 = (1+i)\sqrt{\frac{1}{2}}$. From this we conclude that W is the product of $1+i$ and a positive real quantity, so that necessarily

$$W = (1+i)\sqrt{n}, \quad T = +\sqrt{n}, \quad U = +\sqrt{n}$$

19.

It will be worthwhile to collect all the summations developed so far into a single overview here. In general, we have

$T =$	$U =$	for n of the form
$\pm\sqrt{n}$	$\pm\sqrt{n}$	4μ
$\pm\sqrt{n}$	0	$4\mu+1$
0	0	$4\mu+2$
0	$\pm\sqrt{n}$	$4\mu+3$

and, when we set $k = 1$, the radical must be given the positive sign. What we observed by induction in Article 3 for prime values of n has therefore now been proved with full rigor. Nothing remains except to show how to determine the signs in every case, for any values of k. But before we can tackle this task in full generality, we must first consider more closely the cases where n is a prime number or a power of a prime number.

20.

First let n be an odd prime number. From what we set out in Article 10, it is clear that $W = 1 + 2\Sigma r^a = 1 + 2\Sigma R^{ak}$ if we put $R = \cos\omega + i\sin\omega$, with a, as there, denoting indefinitely all the quadratic residues of n lying between 1 and $n-1$. If we likewise use b to denote indefinitely all the quadratic nonresidues between the same limits, we readily see that all the numbers ak are congruent modulo n either to all the a or to all the b (without regard to order), according as k is a residue or a nonresidue. Thus, in the first case, we have

$$W = 1 + 2\Sigma R^a = 1 + R + R^4 + R^9 + \text{etc.} + R^{(n-1)^2}$$

and hence $W = +\sqrt{n}$ if n is of the form $4\mu+1$, and $W = +i\sqrt{n}$ if n is of the form $4\mu+3$.

In the other case, where k is a nonresidue of n, we have

$$W = 1 + 2\Sigma R^b$$

Since all the a, b evidently make up the complete collection of numbers 1, 2, 3 …, and therefore

$$\Sigma R^a + \Sigma R^b = R + R^2 + R^3 + \text{etc.} + R^{n-1} = -1$$

we obtain

$$W = -1 - 2\Sigma R^a = -(1 + R + R^4 + R^9 + \text{etc.} + R^{(n-1)^2})$$

and hence $W = -\sqrt{n}$ if n is of the form $4\mu + 1$, and $W = -i\sqrt{n}$ if n is of the form $4\mu + 3$.

From this we conclude:

first, if n is of the form $4\mu + 1$ and k is a quadratic residue of n,

$$T = +\sqrt{n}, \quad U = 0$$

second, if n is of the form $4\mu + 1$ and k is a nonresidue of n,

$$T = -\sqrt{n}, \quad U = 0$$

third, if n is of the form $4\mu + 3$ and k is a residue of n,

$$T = 0, \quad U = +\sqrt{n}$$

fourth, if n is of the form $4\mu + 3$ and k is a nonresidue of n,

$$T = 0, \quad U = -\sqrt{n}$$

21.

Second, let n be the square or a higher power of an odd prime number p, and put $n = p^{2\varkappa}q$, with q either $= 1$ or $= p$. Here we should first note that, if λ is any integer not divisible by $p^{\varkappa}$, we have

$$\begin{aligned} & r^{\lambda\lambda} + r^{(\lambda+p^{\varkappa}q)^2} + r^{(\lambda+2p^{\varkappa}q)^2} + r^{(\lambda+3p^{\varkappa}q)^2} \\ & \quad + \text{etc.} + r^{(\lambda+n-p^{\varkappa}q)^3} \\ & = r^{\lambda\lambda}\{1 + r^{2\lambda p^{\varkappa}q} + r^{4\lambda p^{\varkappa}q} \\ & \quad + r^{6\lambda p^{\varkappa}q} + \text{etc.} + r^{2\lambda(n-p^{\varkappa}q)}\} \\ & = \frac{r^{\lambda\lambda}(1 - r^{2\lambda n})}{1 - r^{2\lambda p^{\varkappa}q}} = 0 \end{aligned}$$

It is then easy to see that

$$W = 1 + r^{p^{2\varkappa}} + r^{4p^{2\varkappa}} + r^{9p^{2\varkappa}} + \text{etc.} + r^{(n-p^{\varkappa})^2}$$

For the remaining terms of the series

$$1 + r + r^4 + r^9 + \text{etc.} + r^{(n-1)^2}$$

can be divided into $(p^\varkappa - 1)q$ partial series, each with $p^\varkappa$ terms, whose sums vanish under the transformation just given.

It follows that in the case where $q = 1$, or where n is a power of a prime number with an even exponent, we have

$$W = p^\varkappa = +\sqrt{n}, \text{ and so } T = +\sqrt{n}, \quad U = 0$$

In the other case, where $q = p$, or where n is a power of a prime number with an odd exponent, we shall put $r^{p^{2\varkappa}} = \rho$. Then ρ is a proper root of the equation $x^p - 1 = 0$, namely $\rho = \cos \frac{k}{p} 360^0 + i \sin \frac{k}{p} 360^0$, and we obtain

$$\begin{aligned} W &= 1 + \rho + \rho^4 + \rho^9 + \text{etc.} + \rho^{(p^{\varkappa+1}-1)^2} \\ &= p^\varkappa (1 + \rho + \rho^4 + \rho^9 + \text{etc.} + \rho^{(p-1)^2}) \end{aligned}$$

But the sum of the series $1 + \rho + \rho^4 + \rho^9 + \text{etc.} + \rho^{(p-1)^2}$ is determined by the preceding article. It follows directly that

$$\begin{aligned} W &= \pm\sqrt{n} = T, \quad \text{if } p \text{ is of the form } 4\mu + 1 \\ W &= \pm i\sqrt{n} = iU, \quad \text{if } p \text{ is of the form } 4\mu + 3 \end{aligned}$$

with the positive or negative sign, according as k is a residue or a nonresidue of p.

22.

What we set out in Articles 20 and 21 also readily gives the following proposition, which will be of considerable use to us below. Put

$$W' = 1 + r^h + r^{4h} + r^{9h} + \text{etc.} + r^{h(n-1)^2}$$

with h denoting any integer not divisible by p. Then, in the case where $n = p$, or where n is a power of p with an odd exponent, we have

$$\begin{aligned} W' &= W, \text{ if } h \text{ is a quadratic residue of } p \\ W' &= -W, \text{ if } h \text{ is a quadratic nonresidue of } p \end{aligned}$$

For W' evidently arises from W if, for k, we substitute kh. In the first case, k and kh are alike, and in the second they are unlike, with respect to being residues or nonresidues of p.

But in the case where n is a power of p with an even exponent, we evidently have $W' = +\sqrt{n}$, and therefore always $W' = W$.

23.

In Articles 20, 21 and 22 we considered odd prime numbers and powers of such numbers. The case therefore remains where n is a power of two.

For $n = 2$, we evidently have $W = 1 + r = 0$.

For $n = 4$, we obtain $W = 1 + r + r^4 + r^9 = 2 + 2r$. Thus $W = 2 + 2i$ whenever k is of the form $4\mu + 1$, and $W = 2 - 2i$ whenever k is of the form $4\mu + 3$.

For $n = 8$, we have $W = 1+r+r^4+r^9+r^{16}+r^{25}+r^{36}+r^{49} = 2+4r+2r^4 = 4r$. Hence

$$\begin{aligned}
W &= (1+i)\sqrt{8}, \quad \text{whenever } k \text{ is of the form } 8\mu+1\\
W &= (-1+i)\sqrt{8}, \quad \text{whenever } k \text{ is of the form } 8\mu+3\\
W &= (-1-i)\sqrt{8}, \quad \text{whenever } k \text{ is of the form } 8\mu+5\\
W &= (1-i)\sqrt{8}, \quad \text{whenever } k \text{ is of the form } 8\mu+7
\end{aligned}$$

If n is a higher power of two, put $n = 2^{2\varkappa}q$, with q either $= 1$ or $= 2$, and $\varkappa$ greater than 1. Here we must first observe that, if λ is any integer not divisible by $2^{\varkappa-1}$, we have

$$\begin{aligned}
&r^{\lambda\lambda} + r^{(\lambda+2^{\varkappa}q)^2} + r^{(\lambda+2.2^{\varkappa}q)^2} + r^{(\lambda+3.2^{\varkappa}q)^2}\\
&\quad + \text{etc.} + r^{(\lambda+n-2^{\varkappa}q)^2}\\
&= r^{\lambda\lambda}\{1 + r^{2^{\varkappa+1}\lambda q} + r^{2.2^{\varkappa+1}\lambda q}\\
&\quad + r^{3.2^{\varkappa+1}\lambda q} + \text{etc.} + r^{(2n-2^{\varkappa+1}q)\lambda}\}\\
&= \frac{r^{\lambda\lambda}(1 - r^{2\lambda n})}{1 - r^{2^{\varkappa+1}\lambda q}} = 0
\end{aligned}$$

It is then easy to see that

$$W = 1 + r^{2^{2\varkappa-2}} + r^{4.2^{2\varkappa-2}} + r^{9.2^{2\varkappa-2}} + \text{etc.} + r^{(n-2^{\varkappa-1})^2}$$

Put $r^{2^{2\varkappa-2}} = \rho$. Then ρ is a root of the equation $x^{4q} - 1 = 0$, namely $\rho = \cos\frac{k}{4q}360^0 + i\sin\frac{k}{4q}360^0$; we then obtain

$$\begin{aligned}
W &= 1 + \rho + \rho^4 + \rho^9 + \text{etc.} + \rho^{(2^{\varkappa+1}q-1)^2}\\
&= 2^{\varkappa-1}(1 + \rho + \rho^4 + \rho^9 + \text{etc.} + \rho^{(4q-1)^2})
\end{aligned}$$

But the sum of the series $1 + \rho + \rho^4 + \rho^9 + \text{etc.} + \rho^{(4q-1)^2}$ is determined by what we explained for the cases $n = 4$, $n = 8$. From this we conclude

that in the case where $q = 1$, or where n is a power of the number 4, we have

$$W = (1+i)2^{\varkappa} = (1+i)\sqrt{n}, \quad \text{if } k \text{ is of the form } 4\mu + 1$$
$$W = (1-i)2^{\varkappa} = (1-i)\sqrt{n}, \quad \text{if } k \text{ is of the form } 4\mu + 3$$

these are exactly the formulas given for $n = 4$;

but in the case where $q = 2$, or where n is a power of two with an odd exponent greater than 3, we have

$$W = (1+i)2^{\varkappa}\sqrt{2} = (1+i)\sqrt{n}, \quad \text{if } k \text{ is of the form } 8\mu + 1$$
$$W = (-1+i)2^{\varkappa}\sqrt{2} = (-1+i)\sqrt{n}, \quad \text{if } k \text{ is of the form } 8\mu + 3$$
$$W = (-1-i)2^{\varkappa}\sqrt{2} = (-1-i)\sqrt{n}, \quad \text{if } k \text{ is of the form } 8\mu + 5$$
$$W = (1-i)2^{\varkappa}\sqrt{2} = (1-i)\sqrt{n}, \quad \text{if } k \text{ is of the form } 8\mu + 7$$

these too agree exactly with those we gave for $n = 8$.

24.

Here too it will be worthwhile to determine the ratio of the sum of the series

$$W' = 1 + r^h + r^{4h} + r^{9h} + \text{etc.} + r^{h(n-1)^2}$$

to W, where h denotes any odd integer. Since W' arises from W by changing k into kh, the value of W' depends on the form of the number kh in the same way as W depends on the form of k. Put $\frac{W'}{W} = l$. It is clear that

I. in the case where $n = 4$, or it is a higher power of two with an even exponent, we have

$$l = 1, \quad \text{if } h \text{ is of the form } 4\mu + 1$$
$$l = -i, \quad \text{if } h \text{ is of the form } 4\mu + 3, \text{ and } k \text{ is of the form } 4\mu + 1$$
$$l = +i, \quad \text{if } h \text{ is of the form } 4\mu + 3, \text{ and } k \text{ is of the same form.}$$

II. in the case where $n = 8$, or it is a higher power of two with an odd exponent, we have

$l = 1$, if h is of the form $8\mu + 1$,
$l = -1$, if h is of the form $8\mu + 5$,
$l = +i$, if either h is of the form $8\mu + 3$, and k is of the form $4\mu + 1$,
or h is of the form $8\mu + 7$, and k is of the form $4\mu + 3$,
$l = -i$, if either h is of the form $8\mu + 3$, and k is of the form $4\mu + 3$,
or h is of the form $8\mu + 7$, and k is of the form $4\mu + 1$.

The preceding articles have completely determined the sum W for the cases where n is a prime number or a power of a prime number. It therefore remains to complete the cases where n is composed of several prime numbers; the following theorem will prepare the way for us:

25.

THEOREM. *Let n be the product of two coprime positive integers a, b, and put*

$$P = 1 + r^{aa} + r^{4aa} + r^{9aa} + \text{etc.} + r^{(b-1)^2 aa}$$
$$Q = 1 + r^{bb} + r^{4bb} + r^{9bb} + \text{etc.} + r^{(a-1)^2 bb}$$

Then I claim that $W = PQ$.

Proof. Let α denote indefinitely the numbers $0, 1, 2, 3 \ldots . a - 1$, β the numbers $0, 1, 2, 3 \ldots . b - 1$, and ν the numbers $0, 1, 2, 3 \ldots . n - 1$. Then evidently

$$P = \Sigma r^{aa\beta\beta}, \quad Q = \Sigma r^{bb\alpha\alpha}, \quad W = \Sigma r^{\nu\nu}$$

Thus $PQ = \Sigma r^{aa\beta\beta + bb\alpha\alpha}$, on substituting all the values of α and β, combined with one another in every way. Further, since $2ab\alpha\beta = 2\alpha\beta n$, we have $PQ = \Sigma r^{(a\beta + b\alpha)^2}$. But it is easy to see that the individual values of $a\beta + b\alpha$ are different from one another and equal to some value of ν. Hence $PQ = \Sigma r^{\nu\nu} = W$.

It should also be noted that r^{aa} is a proper root of the equation $x^b - 1 = 0$, and r^{bb} a proper root of the equation $x^a - 1 = 0$.

26.

Now let n be the product of three pairwise coprime numbers a, b, c. It is clear that, on putting $bc = b'$, a and b' will also be coprime; hence W is the product of the two factors

$$1 + r^{aa} + r^{4aa} + r^{9aa} + \text{etc.} + r^{(b'-1)^2 aa}$$
$$1 + r^{b'b'} + r^{4b'b'} + r^{9b'b'} + \text{etc.} + r^{(a-1)^2 b'b'}$$

But since r^{aa} is a proper root of the equation $x^{bc} - 1 = 0$, the first factor itself will be the product of

$$1 + \rho^{bb} + \rho^{4bb} + \rho^{9bb} + \text{etc.} + \rho^{(c-1)^2 bb}$$
$$1 + \rho^{cc} + \rho^{4cc} + \rho^{9cc} + \text{etc.} + \rho^{(b-1)^2 cc}$$

if we put $r^{aa} = \rho$. It is therefore clear that W is the product of the three factors

$$1 + r^{bbcc} + r^{4bbcc} + r^{9bbcc} + \text{etc.} + r^{(a-1)^2 bbcc}$$
$$1 + r^{aacc} + r^{4aacc} + r^{9aacc} + \text{etc.} + r^{(b-1)^2 aacc}$$
$$1 + r^{aabb} + r^{4aabb} + r^{9aabb} + \text{etc.} + r^{(c-1)^2 aabb}$$

where r^{bbcc}, r^{aacc}, r^{aabb} are respectively proper roots of the equations $x^a - 1 = 0$, $x^b - 1 = 0$, $x^c - 1 = 0$.

27.

It is easy to conclude in general that, if n is the product of any number of pairwise coprime factors a, b, c, etc., W is the product of the same number of factors, namely

$$1 + r^{\frac{nn}{aa}} + r^{\frac{4nn}{aa}} + r^{\frac{9nn}{aa}} + \text{etc.} + r^{\frac{(a-1)^2 nn}{aa}}$$
$$1 + r^{\frac{nn}{bb}} + r^{\frac{4nn}{bb}} + r^{\frac{9nn}{bb}} + \text{etc.} + r^{\frac{(b-1)^2 nn}{bb}}$$
$$1 + r^{\frac{nn}{cc}} + r^{\frac{4nn}{cc}} + r^{\frac{9nn}{cc}} + \text{etc.} + r^{\frac{(c-1)^2 nn}{cc}}, \text{ etc.}$$

where $r^{\frac{nn}{aa}}$, $r^{\frac{nn}{bb}}$, $r^{\frac{nn}{cc}}$, etc. are proper roots of the equations $x^a - 1 = 0$, $x^b - 1 = 0$. $x^c - 1 = 0$, etc.

28.

These principles now make the transition to a complete determination of W for any value of n immediately clear. Specifically, resolve n into factors a, b, c, etc., that are either distinct prime numbers or powers of distinct prime numbers, and put $r^{\frac{nn}{aa}} = A$, $r^{\frac{nn}{bb}} = B$, $r^{\frac{nn}{cc}} = C$, etc. Then A, B, C, etc. will be proper roots of the equations $x^a - 1 = 0$, $x^b - 1 = 0$, $x^c - 1 = 0$, etc., and W the product of the factors

$$1 + A + A^4 + A^9 + \text{etc.} + A^{(a-1)^2}$$
$$1 + B + B^4 + B^9 + \text{etc.} + B^{(b-1)^2}$$
$$1 + C + C^4 + C^9 + \text{etc.} + C^{(c-1)^2}$$

Each of these factors can be determined from what we showed in Articles 20, 21, 23, so the value of the product will also become known. It will be useful to collect the rules for determining these factors in a single overview here. Since the root A is $= \frac{kn}{a} \cdot \frac{360^0}{a}$, the sum $1 + A + A^4 + A^9 + \text{etc.} + A^{(a-1)^2}$, which we shall denote by L, will be determined by the number $\frac{kn}{a}$ in the same way as W is determined by k in our general investigation. Twelve cases must now be distinguished.

I. If a is a prime number of the form $4\mu + 1$, say $= p$, or a power of such a prime number with an odd exponent, and $\frac{kn}{a}$ is a quadratic residue of p, then $L = +\sqrt{a}$.

II. If the other conditions remain unchanged and $\frac{kn}{a}$ is a quadratic nonresidue of p, then $L = -\sqrt{a}$.

III. If a is a prime number of the form $4\mu + 3$, say $= p$, or a power of such a prime number with an odd exponent, and $\frac{kn}{a}$ is a quadratic residue of p, then $L = +i\sqrt{a}$.

IV. If, with the other conditions as in III, $\frac{kn}{a}$ is a quadratic nonresidue of p, then $L = -i\sqrt{a}$.

V. If a is the square, or a higher power with an even exponent, of an (odd) prime number, then $L = +\sqrt{a}$.

VI. If $a = 2$, then $L = 0$.

VII. If $a = 4$, or it is a higher power of two with an even exponent, and $\frac{kn}{a}$ is of the form $4\mu + 1$, then $L = (1 + i)\sqrt{a}$.

VIII. If, with the other conditions as in VII, $\frac{kn}{a}$ is of the form $4\mu + 3$, then $L = (1 - i)\sqrt{a}$.

IX. If $a = 8$, or it is a higher power of two with an odd exponent, and $\frac{kn}{a}$ is of the form $8\mu + 1$, then $L = (1 + i)\sqrt{a}$.

X. If, with the other conditions as in IX, $\frac{kn}{a}$ is of the form $8\mu + 3$, then $L = (-1 + i)\sqrt{a}$.

XI. If the other conditions remain unchanged and $\frac{kn}{a}$ is of the form $8\mu + 5$, then $L = (-1 - i)\sqrt{a}$.

XII. If the other conditions remain unchanged and $\frac{kn}{a}$ is of the form $8\mu+7$, then $L = (1-i)\sqrt{a}$.

29.

For example, let $n = 2520 = 8 \,.\, 9 \,.\, 5 \,.\, 7$ and $k = 13$. Here we obtain

for $a = 8$, by case XII, $L = (1-i)\sqrt{8}$
for the factor 9, by case V, the corresponding sum is $= \sqrt{9}$
for the factor 5, by case II, the corresponding sum is $= -\sqrt{5}$
for the factor 7, by case III, the corresponding sum is $= +i\sqrt{7}$.

Thus $W = (1-i) \,.\, (-i) \,.\, \sqrt{2520} = (-1-i)\sqrt{2520}$.
For the same value of n, let $k = 1$; then the corresponding sums are

for the factor 8 the sum $(-1+i)\sqrt{8}$
for the factor 9 the sum $\sqrt{9}$
for the factor 5 the sum $\sqrt{5}$
for the factor 7 the sum $-i\sqrt{7}$

This gives the product $W = (1+i)\sqrt{2520}$.

30.

Another method for determining the sum W in general follows from what we set out in Articles 22. 24. Put $\cos\omega + i\sin\omega = \rho$, and

$$\rho^{\frac{nn}{aa}} = \alpha, \quad \rho^{\frac{nn}{bb}} = \beta, \quad \rho^{\frac{nn}{cc}} = \gamma \text{ etc.}$$

so that $r = \rho^k$, $A = \alpha^k$, $B = \beta^k$, $C = \gamma^k$ etc. Then

$$1 + \rho + \rho^4 + \rho^9 + \text{etc.} + \rho^{(n-1)^2}$$

is the product of the factors

$$1 + \alpha + \alpha^4 + \alpha^9 + \text{etc.} + \alpha^{(a-1)^2}$$
$$1 + \beta + \beta^4 + \beta^9 + \text{etc.} + \beta^{(b-1)^2}$$
$$1 + \gamma + \gamma^4 + \gamma^9 + \text{etc.} + \gamma^{(c-1)^2}, \text{ etc.}$$

and therefore W is the product of the factors

$$w = 1 + \rho + \rho^4 + \rho^9 + \text{etc.} + \rho^{(n-1)^2}$$

$$\mathfrak{A} = \frac{1 + A + A^4 + A^9 + \text{etc.} + A^{(a-1)^2}}{1 + \alpha + \alpha^4 + \alpha^9 + \text{etc.} + \alpha^{(a-1)^2}}$$

$$\mathfrak{B} = \frac{1 + B + B^4 + B^9 + \text{etc.} + B^{(b-1)^2}}{1 + \beta + \beta^4 + \beta^9 + \text{etc.} + \beta^{(b-1)^2}}$$

$$\mathfrak{C} = \frac{1 + C + C^4 + C^9 + \text{etc.} + C^{(c-1)^2}}{1 + \gamma + \gamma^4 + \gamma^9 + \text{etc.} + \gamma^{(c-1)^2}}$$

The first factor w has already been determined by the investigations given above (Article 19); the remaining factors $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}$ etc. follow from the formulas of Articles 22. 24, which we collect here again to have everything together[*]. Twelve cases must be distinguished here, namely

I. If a is an (odd) prime number $= p$, or a power of such a number with an odd exponent, and k is a quadratic residue of p, the corresponding factor is $\mathfrak{A} = +1$.

II. If the other conditions remain unchanged and k is a quadratic non-residue of p, then $\mathfrak{A} = -1$.

III. If a is the square of an odd prime number, or a higher power of it with an even exponent, then $\mathfrak{A} = +1$.

IV. If a is $= 4$, or a higher power of two with an even exponent, and k is of the form $4\mu + 1$, then $\mathfrak{A} = +1$.

V. If, with the other conditions as in IV, k is of the form $4\mu + 3$ and $\frac{n}{a}$ of the form $4\mu + 1$, then $\mathfrak{A} = -i$.

VI. If, with the other conditions as in IV, k is of the form $4\mu + 3$ and $\frac{n}{a}$ of the form $4\mu + 3$, then $\mathfrak{A} = +i$.

VII. If a is $= 8$, or a higher power of two with an odd exponent, and k is of the form $8\mu + 1$, then $\mathfrak{A} = +1$.

VIII. If, with the other conditions as in VII, k is of the form $8\mu + 5$, then $\mathfrak{A} = -1$.

IX. If, with the other conditions as in VII, k is of the form $8\mu + 3$ and $\frac{n}{a}$ of the form $4\mu + 1$, then $\mathfrak{A} = +i$.

X. If, with the other conditions as in VII, k is of the form $8\mu + 3$ and $\frac{n}{a}$ of the form $4\mu + 3$, then $\mathfrak{A} = -i$.

XI. If, with the other conditions as in VII, k is of the form $8\mu + 7$ and $\frac{n}{a}$ of the form $4\mu + 1$, then $\mathfrak{A} = -i$.

XII. If, with the other conditions as in VII, k is of the form $8\mu + 7$ and $\frac{n}{a}$ of the form $4\mu + 3$, then $\mathfrak{A} = +i$.

[*] Clearly, what were k and h there are here $\frac{n}{a}$ and k for the second factor, $\frac{n}{b}$ and k for the third factor, etc.

We omit the case where $a = 2$; here $\mathfrak{A}$ would indeed be $= \frac{0}{0}$, or indeterminate, but then we always have $W = 0$.

The remaining factors $\mathfrak{B}, \mathfrak{C}$ etc. depend on b, c etc. in the same way as $\mathfrak{A}$ depends on a, insofar as these enter into their determination.

31.

By this second method, the first example of Article 29 runs as follows:

The factor w is $= (1+i)\sqrt{2520}$
For $a = 8$ the corresponding factor $\mathfrak{A}$ is, by case VIII, $= -1$
For the second factor of n, 9, the corresponding factor is $+1$ (by case III.)
For the factor 5 the corresponding factor is -1 (by case II.)
For the factor 7 the corresponding factor is -1 (by case II.)

This gives the product $W = (-1-i)\sqrt{2520}$, as in Article 29.

32.

Since the value of W can be determined by *two* methods, one of which rests on the relations of the numbers $\frac{nk}{a}$, $\frac{nk}{b}$, $\frac{nk}{c}$ etc. to the numbers a, b, c etc., while the other depends on the relations of k to the numbers a, b, c etc., there must be a conditional connection among all these relations, so that any one can be determined from the others. Suppose that all the numbers a, b, c etc. are odd prime numbers, and take $k = 1$. Divide the factors a, b, c etc. into two classes: one contains those of the form $4\mu + 1$, denoted by p, p', p'' etc.; the other consists of those of the form $4\mu+3$, denoted by q, q', q'' etc. We shall denote the number of the latter by m. With this done, we first observe that n is of the form $4\mu + 1$ if m is even (this also includes the case where there are no factors at all in the second class, or $m = 0$), whereas n is of the form $4\mu+3$ if m is odd. Now the determination of W by the first method is carried out as follows. Let the numbers $P, P'\ P''$ etc., Q, Q', Q'' etc. depend on the relations of the numbers $\frac{n}{p}$, $\frac{n}{p'}$, $\frac{n}{p''}$ etc., $\frac{n}{q}$, $\frac{n}{q'}$, $\frac{n}{q''}$ etc. to the numbers p, p', p'' etc., q, q', q'' etc., respectively, in such a way that we put

$$P = +1, \text{ if } \frac{n}{p} \text{ is a quadratic residue of } p$$
$$P = -1, \text{ if } \frac{n}{p} \text{ is a quadratic nonresidue of } p$$

and similarly for the others. Then W will be the product of the factors $P\sqrt{p}$, $P'\sqrt{p'}$, $P''\sqrt{p''}$ etc. $iQ\sqrt{q}$, $iQ'\sqrt{q'}$, $iQ''\sqrt{q''}$ etc., and therefore

$$W = PP'P'' \ldots QQ'Q'' \ldots i^m\sqrt{n}$$

By the second method, or rather directly from the rules of Article 19, we have

$$W = +\sqrt{n}, \text{ if } n \text{ is of the form } 4\mu + 1,$$
$$\text{or equivalently, if } m \text{ is even}$$
$$W = +i\sqrt{n}, \text{ if } n \text{ is of the form } 4\mu + 3,$$
$$\text{or if } m \text{ is odd.}$$

Both cases can be combined in the following formula:

$$W = i^{mm}\sqrt{n}$$

It therefore follows that

$$PP'P'' \ldots QQ'Q'' \ldots = i^{mm-m}$$

But i^{mm-m} is $= 1$ whenever m is of the form 4μ or $4\mu + 1$, and $= -1$ whenever m is of the form $4\mu+2$ or $4\mu+3$. From this we deduce the following most elegant

THEOREM. *Let a, b, c etc. denote distinct positive odd prime numbers, and put their product $= n$. Let m of them be of the form $4\mu + 3$, and the rest of the form $4\mu + 1$. The number of those among a, b, c etc. for which $\frac{n}{a}$, $\frac{n}{b}$, $\frac{n}{c}$ etc., respectively, are nonresidues will be even whenever m is of the form 4μ or $4\mu + 1$, but odd whenever m is of the form $4\mu + 2$ or $4\mu + 3$.*

Thus, for example, putting $a = 3$, $b = 5$, $c = 7$, $d = 11$, we have three numbers of the form $4\mu + 3$, namely 3, 7 and 11; and $5 \,.\, 7 \,.\, 11\, R3$; $3 \,.\, 7 \,.\, 11\, R5$; $3 \,.\, 5 \,.\, 11\, R7$; $3 \,.\, 5 \,.\, 7\, N11$, so that only $\frac{n}{d}$ is a nonresidue of d.

33.

The celebrated *fundamental theorem* about quadratic residues is nothing other than a special case of the theorem just developed. Specifically, by limiting the number of the numbers a, b, c etc. to *two*, it is clear that, if just one of them, or neither, is of the form $4\mu + 3$, we must have either both aRb, bRa, or both aNb, bNa; but if both are of the form $4\mu + 3$, one must be a nonresidue of the other, and the latter a residue of the former. Here, then, is a *fourth* proof of this most important theorem, whose first and second proofs we gave in the Disquisitiones Arithmeticae, and the third recently in a separate paper (*Commentt. T. XVI*); we shall later set out two others, again resting on entirely different principles. It is certainly most remarkable that this most beautiful theorem, which at first so stubbornly eluded every attempt, could later be reached by so many paths, worlds apart from one another.

34.

The remaining theorems, which form a kind of supplement to the fundamental theorem, namely those by which we distinguish the prime numbers for which $-1, +2$ and -2 are residues or nonresidues, can also be derived from the same principles. We shall begin with the residue $+2$.

Putting $n = 8a$, with a a prime number, and $k = 1$, the method of Article 28 makes W the product of two factors. One of them will be $+\sqrt{a}$, or $+i\sqrt{a}$, if 8, or equivalently 2, is a quadratic residue of a; but $-\sqrt{a}$ or $-i\sqrt{a}$ if 2 is a nonresidue of a. The second factor is

$$\begin{aligned}
&(1+i)\sqrt{8}, \quad \text{if } a \text{ is of the form } 8\mu+1\\
&(-1+i)\sqrt{8}, \quad \text{if } a \text{ is of the form } 8\mu+3\\
&(-1-i)\sqrt{8}, \quad \text{if } a \text{ is of the form } 8\mu+5\\
&(1-i)\sqrt{8}, \quad \text{if } a \text{ is of the form } 8\mu+7
\end{aligned}$$

But by Article 18 we always have $W = (1+i)\sqrt{n}$; dividing this value by the four values of the second factor shows that the first factor must be

$$\begin{aligned}
&+\sqrt{a}, \quad \text{if } a \text{ is of the form } 8\mu+1\\
&-i\sqrt{a}, \quad \text{if } a \text{ is of the form } 8\mu+3\\
&-\sqrt{a}, \quad \text{if } a \text{ is of the form } 8\mu+5\\
&+i\sqrt{a}, \quad \text{if } a \text{ is of the form } 8\mu+7
\end{aligned}$$

It follows immediately that 2 must be a residue of a in the first and fourth cases, but a nonresidue in the second and third.

35.

The prime numbers for which -1 is a residue or a nonresidue are easily distinguished with the aid of the following theorem, which is also quite remarkable in itself.

THEOREM. *The product of the two factors*

$$W' = 1 + r^{-1} + r^{-4} + \text{etc.} + r^{-(n-1)^2}$$
$$W = 1 + r + r^4 + \text{etc.} + r^{(n-1)^2}$$

is $= n$ *if* n *is odd; or* $= 0$ *if* n *is twice an odd number; or* $= 2n$ *if* n *is divisible by four.*

Proof. Since we evidently have

$$\begin{aligned} W &= r + r^4 + r^9 + \text{etc.} + r^{nn} \\ &= r^4 + r^9 + \text{etc.} + r^{(n+1)^2} \\ &= r^9 + \text{etc.} + r^{(n+2)^2} \quad \text{etc.} \end{aligned}$$

the product WW' can also be written as follows:

$$\begin{aligned} & 1 + r + r^4 + r^9 + \text{etc.} + r^{(n-1)^2} \\ & + r^{-1}(r + r^4 + r^9 + r^{16} + \text{etc.} + r^{nn}) \\ & + r^{-4}(r^4 + r^9 + r^{16} + r^{25} + \text{etc.} + r^{(n+1)^2}) \\ & + r^{-9}(r^9 + r^{16} + r^{25} + r^{36} + \text{etc.} + r^{(n+2)^2}) \\ & \text{etc.} \\ & + r^{-(n-1)^2}(r^{(n-1)^2} + r^{nn} + r^{(n+1)^2} \\ & \qquad + r^{(n+2)^2} + \text{etc.} + r^{(2n-2)^2}) \end{aligned}$$

Summing this expression vertically gives

$$\begin{aligned} & n \\ & + r(1 + rr + r^4 + r^6 + \text{etc.} + r^{2n-2}) \\ & + r^4(1 + r^4 + r^8 + r^{12} + \text{etc.} + r^{4n-4}) \\ & + r^9(1 + r^6 + r^{12} + r^{18} + \text{etc.} + r^{6n-6}) \\ & + \text{etc.} \\ & + r^{(n-1)^2}(1 + r^{2n-2} + r^{4n-4} \\ & \qquad + r^{6n-6} + \text{etc.} + r^{2(n-1)^2}) \end{aligned}$$

Now, if n is odd, each part of this expression, except the first, n, will be $= 0$; for the second is evidently $\frac{r(1-r^{2n})}{1-rr}$, the third $\frac{r^4(1-r^{4n})}{1-r^4}$ etc. Whenever n is even, however, we must also except the part

$$r^{\frac{1}{4}nn}(1 + r^n + r^{2n} + r^{3n} + \text{etc.} + r^{nn-n})$$

which is $= nr^{\frac{1}{4}nn}$. In the first case, therefore, $WW' = n$, but in the second $= n + nr^{\frac{1}{4}nn}$; now $r^{\frac{1}{4}nn}$ is $= +1$ if n is divisible by four, and then we obtain $WW' = 2n$; on the other hand, $r^{\frac{1}{4}nn} = -1$ if n is twice an odd number, in which case $WW' = 0$. Q. E. D.

36.

Article 22 establishes that, if n is an odd prime number, $\frac{W'}{W}$ is $= +1$ or $= -1$ according as -1 is a residue or a nonresidue of n. Hence in the first case we must have $W^2 = +n$, and in the second $W^2 = -n$; by Article 13 we therefore conclude that the first case can occur only when n is of the form $4\mu + 1$, and the second when n is of the form $4\mu + 3$.

Finally, combining the conditions found for the residues $+2$ and -1 immediately shows that -2 is a residue of every prime number of the form $8\mu + 1$ or $8\mu + 3$, and a nonresidue of every prime number of the form $8\mu + 5$ or $8\mu + 7$.

NEW PROOFS AND EXTENSIONS OF THE FUNDAMENTAL THEOREM IN THE THEORY OF QUADRATIC RESIDUES

BY

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NEW PROOFS AND EXTENSIONS OF THE FUNDAMENTAL THEOREM IN THE THEORY OF QUADRATIC RESIDUES.

The fundamental theorem about quadratic residues, which ranks among the most beautiful truths of higher arithmetic, was readily discovered by induction, but proved with far greater difficulty. It often happens in this field that the proofs of the simplest truths, which seem to offer themselves to the investigator through induction of their own accord, lie deeply hidden and can finally be brought to light only after many unsuccessful attempts, perhaps by a path very different from the one along which they were sought. Then it often happens that, once one path has been found, *several* others gradually open up, leading to the same goal: some more briefly and directly, others as if by a roundabout way, starting from very different principles between which and the question at hand one would scarcely have suspected any connection. This remarkable connection among more hidden truths not only gives these investigations a particular charm; it also deserves careful study and unravelling, since it often yields new resources or advances for the science itself.

Although the arithmetical theorem considered here might therefore seem, through the earlier efforts that supplied four entirely different proofs[*)], to have been fully dealt with, I nevertheless return to the same subject and add two further proofs, which will certainly shed new light on it. The first is related to the third in a certain respect, since it starts from the same lemma; it then follows a different path, however, so that it can rightly be regarded as a new proof, which in elegance will seem, if not superior to that third proof, at least by no means inferior. The sixth proof rests on an entirely different and subtler principle, and gives a new example of the remarkable connection between arithmetical truths that at first sight seem very far apart. These two proofs are accompanied by a very simple new algorithm for deciding whether a given integer is a quadratic residue or a nonresidue of a given prime number.

[*)] Two are given in the fourth and fifth sections of the *Disquisitiones Arithmeticae*; the third in a separate paper (*Commentt. Soc. Gotting. Vol. XVI*), and the fourth in the paper *Summatio quarundam serierum singularium* (*Commentt. Recentiores, Vol. I*).

There was another reason that led me to publish now, in particular, the new proofs promised nine years ago. When I began in 1805 to investigate the theory of cubic and biquadratic residues, a far more difficult subject, I experienced much the same course of events as before in the theory of quadratic residues. The theorems that completely settle these questions, and display a remarkable analogy with the theorems about quadratic residues, were immediately discovered by induction as soon as they were sought along a suitable path; but all attempts to obtain proofs complete in every respect remained unsuccessful for a long time. This was precisely what prompted me to strive so eagerly to add more and more proofs to those already known for quadratic residues, sustained by the hope that, among many different methods, one or another might help to illuminate the related subject. This hope was by no means in vain, and tireless work was finally followed by good results. Soon it will be possible to bring the fruits of my nights of work before the public; but before I undertake this arduous task, I have resolved to return once more to the theory of quadratic residues, complete everything that still remains to be done in it, and thus, as it were, take leave of this part of higher arithmetic.

FIFTH PROOF OF THE FUNDAMENTAL THEOREM IN THE THEORY OF QUADRATIC RESIDUES.

1.

We have already stated in the introduction that the fifth and third proofs start from the same lemma. For convenience, we have thought it appropriate to repeat it here in notation adapted to the present investigation.

LEMMA. *Let m be a prime number (positive and odd), and M an integer not divisible by m. Take the least positive residues of the numbers*

$$M,\ 2M,\ 3M,\ 4M\ \ldots\ldots\ \tfrac{1}{2}(m-1)M$$

modulo m. Some will be less than $\frac{1}{2}m$, and others greater; let the number of the latter be $= n$. Then M is a quadratic residue or a nonresidue of m, according as n is even or odd.

PROOF. Let those of these residues which are less than $\frac{1}{2}m$ be a, b, c, d etc., and the remaining ones, greater than $\frac{1}{2}m$, be a', b', c', d' etc. The complements of the latter to m, namely $m-a'$, $m-b'$, $m-c'$, $m-d'$ etc., are evidently all less than $\frac{1}{2}m$ and distinct both from one another and from the residues a, b, c, d etc. Thus, taken together with these, they are precisely all the numbers $1,\ 2,\ 3,\ 4\ldots.\frac{1}{2}(m-1)$, though in a different order. Putting the product

$$1\,.\,2\,.\,3\,.\,4\ldots..\tfrac{1}{2}(m-1) = P$$

we therefore have

$$P = abcd\ldots. \times (m-a')(m-b')(m-c')(m-d')\ldots.$$

and hence

$$(-1)^n P = abcd\ldots. \times (a'-m)(b'-m)(c'-m)(d'-m)\ldots.$$

Furthermore, modulo m, we have

$$\begin{aligned} PM^{\frac{1}{2}(m-1)} &\equiv abcd\,..\times a'b'c'd'\,..\\ &\equiv abcd\,..\times (a'-m)(b'-m)(c'-m)(d'-m)\,.. \end{aligned}$$

and hence

$$PM^{\frac{1}{2}(m-1)} \equiv P(-1)^n$$

It follows that $M^{\frac{1}{2}(m-1)} \equiv \pm 1$, taking the upper or lower sign according as n is even or odd. With the aid of the theorem proved in Article 106 of the *Disquisitiones Arithmeticae*, the lemma follows immediately.

2.

THEOREM. *Let m, M be positive odd integers coprime to one another, and let n be the number of those among the least positive residues of the numbers*

$$M,\ 2M,\ 3M \ldots\ldots \tfrac{1}{2}(m-1)M$$

modulo m which are greater than $\frac{1}{2}m$. Likewise, let N be the number of those among the least positive residues of the numbers

$$m,\ 2m,\ 3m \ldots\ldots \tfrac{1}{2}(M-1)m$$

modulo M which are greater than $\frac{1}{2}M$. Then the three numbers n, N, $\frac{1}{4}(m-1)(M-1)$ are either all even, or one is even and the other two odd.

PROOF. Let us denote

by f the collection of numbers $1,\ 2,\ 3 \ldots\ldots \frac{1}{2}(m-1)$.
by f' the collection of numbers $m-1,\ m-2,\ m-3 \ldots\ldots \frac{1}{2}(m+1)$
by F the collection of numbers $1,\ 2,\ 3 \ldots\ldots \frac{1}{2}(M-1)$
by F' the collection of numbers $M-1,\ M-2,\ M-3 \ldots\ldots \frac{1}{2}(M+1)$

Thus n indicates how many numbers Mf have their least positive residues modulo m in the collection f'. Likewise, N indicates how many numbers mF have their least positive residues modulo M in the collection F'. Finally, let

φ denote the collection of numbers $1,\ 2,\ 3 \ldots\ldots \frac{1}{2}(mM-1)$
φ' denote the collection of numbers $mM-1,\ mM-2,\ mM-3 \ldots\ldots \frac{1}{2}(mM+1)$

Every integer not divisible by m must be congruent modulo m either to a residue from f or to one from f'. Likewise, every integer not divisible by M is congruent modulo M either to a residue from F or to one from F'. Thus all the numbers in φ, among which evidently none is divisible by both m and M, can be divided into eight classes as follows.

I. The first class contains the numbers congruent modulo m to a number from f, and modulo M to a number from F. We shall denote their number by α.

II. The numbers congruent modulo m, M, respectively, to numbers from f, F'; we put their number $= \beta$.

III. The numbers congruent modulo m, M, respectively, to numbers from f', F; we put their number $= \gamma$.

IV. The numbers congruent modulo m, M, respectively, to numbers from f', F'; let their number be $= \delta$.

V. The numbers divisible by m, but congruent modulo M to residues from F.

VI. The numbers divisible by m, but congruent modulo M to residues from F'.

VII. The numbers divisible by M, but congruent modulo m to residues from f.

VIII. The numbers divisible by M, but congruent modulo m to residues from f'.

Evidently classes V and VI together contain all the numbers mF. The number contained in VI is $= N$, so the number contained in V is $\frac{1}{2}(M-1)-N$. Likewise, classes VII and VIII together contain all the numbers Mf; class VIII contains n numbers, and class VII contains $\frac{1}{2}(m-1)-n$.

In just the same way, all the numbers in φ' are divided into eight classes IX . . XVI. If we keep the same order in doing this, it is easy to see that the numbers in the classes

$$\text{IX, X, XI, XII, XIII, XIV, XV, XVI}$$

are, respectively, the complements to mM of the numbers in the classes

$$\text{IV, III, II, I, VI, V, VIII, VII}$$

so that class IX contains δ numbers, class X contains γ, and so on. It is now clear that, if all the numbers of the first class are combined with all those of the ninth, we obtain all the numbers below mM which are congruent modulo m to a number from f, and modulo M to a number from F. It is easy to see that their number equals the number of all combinations of each individual f with each individual F. We therefore have

$$\alpha + \delta = \tfrac{1}{4}(m-1)(M-1)$$

and by a similar argument we also have

$$\beta + \gamma = \tfrac{1}{4}(m-1)(M-1)$$

Combining all the numbers of classes II, IV and VI evidently gives all the numbers below $\frac{1}{2}mM$ which are congruent to a residue from F' modulo M. These same numbers can also be written as follows:

$$F',\ M+F',\ 2M+F',\ 3M+F' \ldots .\ \tfrac{1}{2}(m-3)M+F'$$

Their total number is therefore $= \frac{1}{4}(m-1)(M-1)$, or we have

$$\beta + \delta + N = \tfrac{1}{4}(m-1)(M-1)$$

Likewise, combining all of classes III, IV and VIII gives

$$\gamma + \delta + n = \tfrac{1}{4}(m-1)(M-1)$$

These four equations give the following:

$$2\alpha = \tfrac{1}{4}(m-1)(M-1) + n + N$$
$$2\beta = \tfrac{1}{4}(m-1)(M-1) + n - N$$
$$2\gamma = \tfrac{1}{4}(m-1)(M-1) - n + N$$
$$2\delta = \tfrac{1}{4}(m-1)(M-1) - n - N$$

each of which establishes the theorem.

3.

If we now suppose that m and M are prime numbers, the fundamental theorem follows immediately by combining the preceding theorem with the lemma of Article 1. For it is clear that

I. whenever the numbers m, M both have the form $4k+1$, or just one of them does, the number $\frac{1}{4}(m-1)(M-1)$ is even. Thus n and N are either both even or both odd, and consequently either each of m and M is a quadratic residue of the other, or each is a quadratic nonresidue of the other.

II. Whenever both numbers m, M are of the form $4k+3$, however, $\frac{1}{4}(m-1)(M-1)$ is odd. Hence one of the numbers n, N is even and the other odd, and consequently one of the numbers m, M is a quadratic residue of the other, and the other a quadratic nonresidue of the first. Q. E. D.

SIXTH PROOF OF THE FUNDAMENTAL THEOREM IN THE THEORY OF QUADRATIC RESIDUES.

1.

THEOREM. *Let p denote a prime number (positive and odd), n a positive integer not divisible by p, and x an indeterminate quantity. Then the function*

$$1 + x^n + x^{2n} + x^{3n} + \text{etc.} + x^{np-n}$$

is divisible by

$$1 + x + xx + x^3 + \text{etc.} + x^{p-1}$$

PROOF. Take a positive integer g such that $gn \equiv 1 \pmod{p}$, and put $gn = 1 + hp$. Then we have

$$\begin{aligned}\frac{1 + x^n + x^{2n} + x^{3n} + \text{etc.}\, x^{np-n}}{1 + x + xx + x^3 + \text{etc.}\, x^{p-1}} &= \frac{(1 - x^{np})(1 - x)}{(1 - x^n)(1 - x^p)} \\ &= \frac{(1 - x^{np})(1 - x^{gn} - x + x^{hp+1})}{(1 - x^n)(1 - x^p)} \\ &= \frac{1 - x^{np}}{1 - x^p} \cdot \frac{1 - x^{gn}}{1 - x^n} - \frac{x(1 - x^{np})}{1 - x^n} \cdot \frac{1 - x^{hp}}{1 - x^p}\end{aligned}$$

which is evidently a polynomial. Q. E. D.

Thus every polynomial in x divisible by $\frac{1-x^{np}}{1-x^n}$ is also divisible by $\frac{1-x^p}{1-x}$.

2.

Let α denote a positive primitive root modulo p; that is, let α be a positive integer such that the least positive residues of the powers $1, \alpha, \alpha\alpha, \alpha^3 \ldots\ldots \alpha^{p-2}$ modulo p are precisely the numbers $1, 2, 3, 4 \ldots\ldots p-1$, without regard to order. Further, denoting by fx the function

$$x + x^\alpha + x^{\alpha\alpha} + x^{\alpha^3} + \text{etc.} + x^{\alpha^{p-2}} + 1$$

it is clear that $fx - 1 - x - xx - x^3 - \text{etc.} - x^{p-1}$ is divisible by $1 - x^p$, and therefore all the more by $\frac{1-x^p}{1-x} = 1 + x + xx + x^3 + \text{etc.} + x^{p-1}$. This last function therefore also divides fx itself. Since x represents an indeterminate quantity, it follows that $f(x^n)$ is also divisible by $\frac{1-x^{np}}{1-x^n}$, and hence, by the preceding article, by $\frac{1-x^p}{1-x}$, whenever n is an integer not divisible by p. If, on the other hand, n is an integer divisible by p, the individual terms of the function $f(x^n)$, each decreased by one, are divisible by $1 - x^p$. In this case, therefore, $f(x^n) - p$ is also divisible by $1 - x^p$, and hence by $\frac{1-x^p}{1-x}$.

3.

THEOREM. *Putting*

$$x - x^{\alpha} + x^{\alpha\alpha} - x^{\alpha^3} + x^{\alpha^4} - \text{etc.} - x^{\alpha^{p-2}} = \xi$$

then $\xi\xi \mp p$ *is divisible by* $\frac{1-x^p}{1-x}$, *taking the upper sign whenever* p *is of the form* $4k+1$, *and the lower whenever* p *is of the form* $4k+3$.

PROOF. It is easy to see that, of the following $p-1$ functions

$$\begin{aligned}
&+ x\xi - xx + x^{\alpha+1} - x^{\alpha\alpha+1} \\
&\qquad + \text{etc.} + x^{\alpha^{p-2}+1} \\
&- x^{\alpha}\xi - x^{2\alpha} + x^{\alpha\alpha+\alpha} - x^{\alpha^3+\alpha} \\
&\qquad + \text{etc.} + x^{\alpha^{p-1}+\alpha} \\
&+ x^{\alpha\alpha}\xi - x^{2\alpha\alpha} + x^{\alpha^3+\alpha\alpha} - x^{\alpha^4+\alpha\alpha} \\
&\qquad + \text{etc.} + x^{\alpha^p+\alpha\alpha} \\
&- x^{\alpha^3}\xi - x^{2\alpha^3} + x^{\alpha^4+\alpha^3} - x^{\alpha^5+\alpha^3} \\
&\qquad + \text{etc.} + x^{\alpha^{p+1}+\alpha^3}
\end{aligned}$$

etc., as far as

$$\begin{aligned}
&- x^{\alpha^{p-2}}\xi - x^{2\alpha^{p-2}} + x^{\alpha^{p-1}+\alpha^{p-2}} - x^{\alpha^p+\alpha^{p-2}} \\
&\qquad\qquad + \text{etc.} + x^{\alpha^{2p-4}+\alpha^{p-2}}
\end{aligned}$$

the first is $= 0$, while each of the others is divisible by $1 - x^p$. Their sum is therefore also divisible by $1 - x^p$, and is found to be

$$\begin{aligned}
&= \xi\xi - (f(xx) - 1) + (f(x^{\alpha+1}) - 1) - (f(x^{\alpha\alpha+1}) - 1) \\
&\qquad + (f(x^{\alpha^3+1}) - 1) - \text{etc.} \\
&\qquad + (f(x^{\alpha^{p-2}+1}) - 1) \\
&= \xi\xi - f(xx) + f(x^{\alpha+1}) - f(x^{\alpha\alpha+1}) \\
&\qquad + f(x^{\alpha^3+1}) - \text{etc.} + f(x^{\alpha^{p-2}+1}) = \Omega
\end{aligned}$$

Thus this expression Ω is also divisible by $\frac{1-x^p}{1-x}$. Now, among the exponents $2,\ \alpha+1,\ \alpha\alpha+1,\ \alpha^3+1 \ldots\ldots \alpha^{p-2}+1$, only one is divisible by p, namely $\alpha^{\frac{1}{2}(p-1)}+1$. Hence, by the preceding article, the following individual parts of the expression Ω

$$f(xx), \quad f(x^{\alpha+1}), \quad f(x^{\alpha\alpha+1}), \quad (fx^{\alpha^3+1}) \quad \text{etc.}$$

with the sole exception of the term $f(x^{\alpha^{\frac{1}{2}(p-1)}+1})$, are divisible by $\frac{1-x^p}{1-x}$. Those parts may therefore be removed, leaving the following function still divisible by $\frac{1-x^p}{1-x}$:

$$\xi\xi \mp f(x^{\alpha^{\frac{1}{2}(p-1)}+1})$$

Here the upper or lower sign applies according as p is of the form $4k+1$ or of the form $4k+3$. Since, moreover, $f(x^{\alpha^{\frac{1}{2}(p-1)}+1})-p$ is divisible by $\frac{1-x^p}{1-x}$, $\xi\xi \mp p$ is also divisible by $\frac{1-x^p}{1-x}$. Q. E. D.

To prevent any ambiguity from the double sign, we shall use ε to denote the number $+1$ or -1, according as p is of the form $4k+1$ or $4k+3$. Thus $\frac{(1-x)(\xi\xi-\varepsilon p)}{1-x^p}$ is a polynomial in x, which we shall denote by Z.

4.

Let q be a positive odd number, so that $\frac{1}{2}(q-1)$ is an integer. Then $(\xi\xi)^{\frac{1}{2}(q-1)} - (\varepsilon p)^{\frac{1}{2}(q-1)}$ is divisible by $\xi\xi - \varepsilon p$, and hence also by $\frac{1-x^p}{1-x}$. Let us put $\varepsilon^{\frac{1}{2}(q-1)} = \delta$, and

$$\xi^{q-1} - \delta p^{\frac{1}{2}(q-1)} = \frac{1-x^p}{1-x} \cdot Y$$

Then Y is a polynomial in x, and $\delta = .+1$ whenever one of the numbers p, q, or indeed both, is of the form $4k+1$. On the other hand, $\delta = -1$ whenever both p, q are of the form $4k+3$.

5.

Let us now suppose that q too is a prime number, different from p. It is clear from the theorem proved in Article 51 of the *Disquisitiones Arithmeticae* that

$$. \quad \xi^q - (x^q - x^{q\alpha} + x^{q\alpha\alpha} - x^{q\alpha^3} + \text{etc.} - x^{q\alpha^{p-2}})$$

is divisible by q, or is of the form qX, where X is a polynomial in x whose numerical coefficients are also integers (as is also to be understood of the other polynomials Z, Y, W occurring here). For the modulus p and primitive root α, let us denote the index of the number q by μ; that is, let $q \equiv \alpha^\mu \pmod{p}$. The numbers $q,\ q\alpha,\ q\alpha\alpha,\ q\alpha^3 \ldots\ldots q\alpha^{p-2}$ are therefore congruent modulo p, respectively, to the numbers $\alpha^\mu,\ \alpha^{\mu+1},\ \alpha^{\mu+2} \ldots\ldots \alpha^{p-2},\ 1,\ \alpha,\ \alpha\alpha \ldots\ldots \alpha^{\mu-1}$, so that

$$\begin{array}{l}
x^q - x^{\alpha^\mu} \\
x^{q\alpha} - x^{\alpha^{1+1}} \\
x^{q\alpha\alpha} - x^{\alpha^{\mu+2}} \\
x^{q\alpha^3} - x^{\alpha^{\mu+3}} \\
\quad . \\
\quad . \\
\quad . \\
\quad . \\
x^{q\alpha^{p-\mu-2}} - x^{\alpha^{p-2}} \\
x^{q\alpha^{p-\mu-1}} - x \\
x^{q\alpha^{p-\mu}} - x^\alpha \\
x^{q\alpha^{p-\mu+1}} - x^{\alpha\alpha} \\
\quad . \\
\quad . \\
\quad . \\
\quad . \\
x^{q\alpha^{p-2}} - x^{\alpha^{\mu-1}}
\end{array}$$

are divisible by $1 - x^p$. Taking these quantities alternately with positive and negative signs and adding them shows that $1 - x^p$ divides the function

$$x^q - x^{q\alpha} + x^{q\alpha\alpha} - x^{q\alpha^3} + \text{etc.} - x^{q\alpha^{p-2}} \mp \xi$$

with the upper or lower sign applying according as μ is even or odd; that is, according as q is a quadratic residue or a nonresidue of p. We shall therefore put

$$x^q - x^{q\alpha} + x^{q\alpha\alpha} - x^{q\alpha^3} + \text{etc.} - x^{q\alpha^{p-2}} - \gamma\xi = (1 - x^p)W$$

taking $\gamma = +1$ or $\gamma = -1$, according as q is a quadratic residue or a nonresidue of p. It is clear that W is a polynomial.

6.

With these preparations, we combine the preceding equations to obtain

$$\begin{aligned} q\xi X = \varepsilon p(\delta p^{\frac{1}{2}(q-1)} - \gamma) & \\ & + \frac{1 - x^p}{1 - x} \cdot (Z(\delta p^{\frac{1}{2}(q-1)} - \gamma) \\ & + Y\xi\xi - W\xi(1 - x)) \end{aligned}$$

Let us suppose that dividing the function ξX by

$$x^{p-1} + x^{p-2} + x^{p-3} + \text{etc.} + x + 1$$

yields the quotient U with remainder T, or that we have

$$\xi X = \frac{1 - x^p}{1 - x} \cdot U + T$$

where U, T are polynomials whose numerical coefficients are also integers, and T is certainly of lower degree than the divisor. We therefore have

$$\begin{aligned} qT - \varepsilon p(\delta p^{\frac{1}{2}(q-1)} - \gamma) & \\ & = \frac{1 - x^p}{1 - x} \cdot (Z(\delta p^{\frac{1}{2}(q-1)} - \gamma) \\ & \quad + Y\xi\xi - W\xi(1 - x) - qU) \end{aligned}$$

This equation evidently cannot hold unless the left side and the right side each vanish separately. Thus $\varepsilon p(\delta p^{\frac{1}{2}(q-1)} - \gamma)$ is divisible by q, as is $\delta p^{\frac{1}{2}(q-1)} - \gamma$; and therefore, since $\delta\delta = 1$, the number $p^{\frac{1}{2}(q-1)} - \gamma\delta$ is also divisible by q.

Now, if β denotes unity taken positively or negatively, according as p is a quadratic residue or a quadratic nonresidue of the number q, then $p^{\frac{1}{2}(q-1)} - \beta$ is divisible by q, and hence so is $\beta - \gamma\delta$. This cannot happen unless $\beta = \gamma\delta$. The fundamental theorem follows immediately. Namely,

I. Whenever the numbers p, q both have the form $4k + 1$, or just one of them does, so that $\delta = +1$, we have $\beta = \gamma$. Consequently either q is a quadratic residue of p and at the same time p a quadratic residue of q; or q is a nonresidue of p and at the same time p a nonresidue of q.

II. Whenever both p, q are of the form $4k + 3$, so that $\delta = -1$, we have $\beta = -\gamma$. Consequently either q is a quadratic residue of p and at the same time p a nonresidue of q; or q is a nonresidue of p and at the same time p a residue of q.

Q. E. D.

A new algorithm for deciding whether a given positive integer is a quadratic residue or a nonresidue of a given positive prime number.

1.

Before presenting a new solution to this problem, we shall briefly recall the solution given in the *Disquisitiones Arithmeticae*. It can be carried out quite readily with the help of the fundamental theorem and the following known theorems:

I. The relation of the number a to the number b (whether the former is a quadratic residue or a nonresidue of the latter) is the same as that of the number c to b, if $a \equiv c \pmod{b}$.

II. If a is the product of the factors $\alpha, \beta, \gamma, \delta$ etc., and b is prime, the relation of a to b depends on the relations of these factors to b as follows: a is a quadratic residue of b or a nonresidue according as an even or an odd number of those factors are nonresidues of b. Thus a factor that is a square can be disregarded entirely in this examination; if a factor is an integer raised to an odd power, that integer itself can take its place.

III. The number 2 is a quadratic residue of every prime of the form $8m+1$ or $8m+7$, and a nonresidue of every prime of the form $8m+3$ or $8m+5$.

Suppose, then, that a number a is given whose relation to a prime b is sought. If a is greater than b, first replace it by its least positive residue modulo b. Once this residue has been resolved into prime factors, theorem II reduces the question to finding the relation of each factor to b. The relation of the factor 2 (if it occurs once, three times, five times, etc.) is known from theorem III; by the fundamental theorem, the relation of each remaining factor depends on the relation of b itself to that factor. Thus, in place of the single relation of the given number to the prime b, we must now investigate several relations of the number b to other odd primes smaller than b. These problems are reduced in the same way to smaller moduli, and these successive reductions must evidently come to an end.

2.

To illustrate this solution, suppose that the relation of the number 103 to 379 is sought. Since 103 is already less than 379 and is itself prime, the fundamental theorem applies immediately: it tells us that the required relation is opposite to that of the number 379 to 103. This in turn is the same as the relation of the number 70 to 103, which itself depends on the relations of 2, 5, 7 to 103. The first of these is known from theorem III. By the fundamental theorem, the second depends on the relation of the number 103 to 5, which by theorem I is the same as that of 3 to 5; this in turn depends, by the fundamental theorem, on the relation of 5 to 3, which by theorem I is the same as that of 2 to 3, known from theorem III. Likewise, by the

fundamental theorem, the relation of 7 to 103 depends on that of 103 to 7, which by theorem I is the same as that of 5 to 7; this in turn depends, by the fundamental theorem, on the relation of 7 to 5, which by theorem I is the same as that of 2 to 5, known from theorem III. If we now wish to turn this analysis into a synthesis, the decision is reduced to fourteen steps. We give them all here, so that the greater elegance of the new solution may stand out more clearly.

1. The number 2 is a quadratic residue of the number 103 (theorem III).
2. The number 2 is a quadratic nonresidue of the number 3 (theorem III).
3. The number 5 is a quadratic nonresidue of the number 3 (from I and 2).
4. The number 3 is a quadratic nonresidue of the number 5 (the fundamental theorem and 3).
5. The number 103 is a quadratic nonresidue of the number 5 (I and 4).
6. The number 5 is a quadratic nonresidue of the number 103 (the fundamental theorem and 5).
7. The number 2 is a quadratic nonresidue of the number 5 (theorem III).
8. The number 7 is a quadratic nonresidue of the number 5 (I and 7).
9. The number 5 is a quadratic nonresidue of the number 7 (the fundamental theorem and 8).
10. The number 103 is a quadratic nonresidue of the number 7 (I and 9).
11. The number 7 is a quadratic residue of the number 103 (the fundamental theorem and 10).
12. The number 70 is a quadratic nonresidue of the number 103 (II, 1, 6, 11).
13. The number 379 is a quadratic nonresidue of the number 103 (I and 12).
14. The number 103 is a quadratic residue of the number 379 (the fundamental theorem and 13).

For brevity, in what follows we shall use the notation introduced in *Comment. Gotting. Vol. XVI.* By $[x]$ we shall mean the quantity x itself when x is an integer, and the greatest integer below x when x is not an integer. Thus $x - [x]$ is always nonnegative and less than unity.

3.

Problem. *Let a, b be positive integers coprime to one another, and put $[\frac{1}{2}a] = a'$. Find the sum*

$$\left[\frac{b}{a}\right] + \left[\frac{2b}{a}\right] + \left[\frac{3b}{a}\right] + \left[\frac{4b}{a}\right] + \text{etc.} + \left[\frac{a'b}{a}\right]$$

SOLUTION. For brevity, let us denote such a sum by $\varphi(a, b)$, so that we also have

$$\varphi(b, a) = \left[\frac{a}{b}\right] + \left[\frac{2a}{b}\right] + \left[\frac{3a}{b}\right] + \text{etc.} + \left[\frac{b'a}{b}\right]$$

if we put $[\frac{1}{2}b] = b'$. In the third proof of the fundamental theorem it was shown that, when a and b are odd,

$$\varphi(a, b) + \varphi(b, a) = a'b'$$

By following the same method, the truth of this proposition extends easily to the case in which one of the numbers $a,\ b$ is odd, as we already indicated there. Divide a by b, as in the method for finding the greatest common divisor of two integers, and let β be the quotient and c the remainder; then divide b by c, and so on, obtaining the equations

$$\begin{aligned} a &= \beta b + c \\ b &= \gamma c + d \\ c &= \delta d + e \\ d &= \varepsilon e + f \text{ etc.} \end{aligned}$$

In this way the continually decreasing sequence of numbers b, c, d, e, f etc. will eventually reach unity, since by hypothesis a and b are coprime. The last equation is therefore

$$k = \lambda l + 1$$

Since evidently

$$\begin{aligned} \left[\frac{a}{b}\right] &= \left[\beta + \frac{c}{b}\right] = \beta + \left[\frac{c}{b}\right] \\ \left[\frac{2a}{b}\right] &= \left[2\beta + \frac{2c}{b}\right] = 2\beta + \left[\frac{2c}{b}\right] \\ \left[\frac{3a}{b}\right] &= \left[3\beta + \frac{3c}{b}\right] = 3\beta + \left[\frac{3c}{b}\right] \end{aligned}$$

etc.. we have

$$\varphi(b, a) = \varphi(b, c) + \tfrac{1}{2}\beta(b'b' + b')$$

and consequently

$$\varphi(a, b) = a'b' - \tfrac{1}{2}\beta(b'b' + b') - \varphi(b, c)$$

By similar reasoning, on putting $[\frac{1}{2}c] = c'$, $[\frac{1}{2}d] = d'$ $[\frac{1}{2}e] = e'$ etc., we obtain

$$\begin{aligned}\varphi(b,c) &= b'c' - \tfrac{1}{2}\gamma(c'c' + c') - \varphi(c,d)\\ \varphi(c,d) &= c'd' - \tfrac{1}{2}\delta(d'd' + d') - \varphi(d,e).\\ \varphi(d,e) &= d'e' - \tfrac{1}{2}\varepsilon(e'e' + e') - \varphi(e,f)\end{aligned}$$

etc., as far as

$$\varphi(k,l) = k'l' - \tfrac{1}{2}\lambda(l'l' + l') - \varphi(l,1)$$

Since evidently $\varphi(l,1) = 0$, we obtain from this the formula

$$\begin{aligned}\varphi(a,b) = {} & a'b' - b'c' + c'd' - d'e' + \text{etc.} \pm k'l'\\ & - \tfrac{1}{2}\beta(b'b' + b') + \tfrac{1}{2}\gamma(c'c' + c') - \tfrac{1}{2}\delta(d'd' + d')\\ & + \tfrac{1}{2}\varepsilon(e'e' + e') - \text{etc.} \mp \tfrac{1}{2}\lambda(l'l' + l')\end{aligned}$$

4.

It now follows easily from the third proof that the relation of the number b to a, whenever a is prime, can be read directly from the value of the sum $\varphi(a, 2b)$. According as this sum is even or odd, b is a quadratic residue of a or a nonresidue. The sum $\varphi(a,b)$ itself can also be used for the same purpose, provided that we distinguish the case in which b is odd from that in which it is even. Namely,

I. When b is odd, b is a quadratic residue or a nonresidue of a according as $\varphi(a,b)$ is even or odd.

II. When b is even, the same rule holds if, in addition, a is of the form $8n+1$ or $8n+7$; but if, for an even value of b, the modulus a is of the form $8n+3$ or $8n+5$, the opposite rule must be applied: b is a quadratic residue of a if $\varphi(a,b)$ is odd, and a nonresidue if $\varphi(a,b)$ is even.

All this follows very easily from Article 4 of the third proof.

5.

Example. To find the relation of the number 103 to the prime 379, we have, for calculating the sum $\varphi(379, 103)$,

$$\begin{array}{r|r|l} a = 379 & a' = 189 & \\ b = 103 & b' = 51 & \beta = 3 \\ c = 70 & c' = 35 & \gamma = 1 \\ d = 33 & d' = 16 & \delta = 2 \\ e = 4 & e' = 2 & \varepsilon = 8 \end{array}$$

and hence

$$\begin{aligned} \varphi(379, 103) &= 9639 - 1785 + 560 - 32 - 3978 + 630 - 272 + 24 \\ &= 4786 \end{aligned}$$

so 103 is a quadratic residue of the number 379. If we prefer to use the sum $(379, 206)$ for the same purpose, we have the following scheme:

$$\begin{array}{r|r|l} 379 & 189 & \\ 206 & 103 & 1 \\ 173 & 86 & 1 \\ 33 & 16 & 5 \\ 8 & 4 & 4 \end{array}$$

from which we obtain

$$\begin{aligned} \varphi(379, 206) &= 19467 - 8858 + 1376 - 64 - 5356 + 3741 - 680 + 40 \\ &= 9666 \end{aligned}$$

Thus 103 is a quadratic residue of the number 379.

6.

To determine the relation of the number b to a, it is not necessary to calculate the individual terms of the sum $\varphi(a, b)$; it is enough to know how many of them are odd. Our rule can therefore also be stated as follows:

As above, take $a = \beta b + c,\ b = \gamma c + d,\ c = \delta d + e$ etc., until the sequence of numbers $a,\ b,\ c,\ d,\ e$ etc. reaches unity. Put $[\frac{1}{2}a] = a',\ [\frac{1}{2}b] = b',\ [\frac{1}{2}c] = c'$ etc., and let μ count the odd numbers in the sequence a', b', c' etc. that are immediately followed by an odd number. Further, let ν count the odd numbers in the sequence β, γ, δ etc. whose corresponding numbers in the sequence b', c', d' etc., respectively, have the form $4n + 1$ or $4n + 2$. Then b is a quadratic residue or a nonresidue of a according as $\mu + \nu$ is even or odd, with the sole exception that, when b is even and a is of the form $8n + 3$ or $8n + 5$, the opposite rule holds.

In our example the sequence a', b', c', d', e' has two instances in which an odd entry is immediately followed by an odd entry, so $\mu = 2$; the sequence $\beta', \gamma', \delta', \varepsilon'$ does contain two odd numbers, but their corresponding numbers in the sequence b', c', d', e' have the form $4n+3$, so $\nu = 0$. Thus $\mu+\nu$ is even, and consequently 103 is a quadratic residue of the number 379.

THEORY

OF BIQUADRATIC RESIDUES

FIRST PAPER

BY

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THEORY OF BIQUADRATIC RESIDUES.

FIRST PAPER.

1.

The theory of quadratic residues reduces to a few fundamental theorems that rank among the most beautiful jewels of higher arithmetic. As is well known, these were first easily discovered through inductive observation and then proved in many different ways, leaving nothing further to be desired.

The theory of cubic and biquadratic residues, however, requires much deeper investigation. When we began to explore it in 1805, besides the results that lie, as it were, at the threshold, some special theorems presented themselves, remarkable both for their simplicity and for the difficulty of their proofs. But we soon found that the principles of arithmetic used up to then were wholly insufficient to establish the general theory. Rather, the theory necessarily requires the field of higher arithmetic to be enlarged, as it were, infinitely many times over. What this means will become entirely clear as these investigations continue. As soon as we entered this new field, inductive observation immediately opened a path to knowledge of the simplest theorems encompassing the entire theory. Their proofs, however, lay so deeply hidden that only after many unsuccessful attempts could they finally be brought to light.

Now that we are preparing to publish these studies, we shall begin with the theory of biquadratic residues. In this first paper we shall present the investigations that could already be completed on this side of the enlarged field of arithmetic. They pave the way to that field, as it were, and also contribute some new developments to the theory of the division of the circle.

2.

We introduced the notion of a biquadratic residue in Article 115 of the *Disquisitiones Arithmeticae*: an integer a, positive or negative, is called a biquadratic residue of the integer p if a can be congruent to a fourth power modulo p, and a biquadratic nonresidue if no such congruence exists. Throughout the following investigations, unless we expressly state otherwise, we shall assume that the modulus p is an odd positive prime and that a is not divisible by p, since all the remaining cases can very easily be reduced to this one.

3.

Clearly, every biquadratic residue of p is also a quadratic residue of the same number, and hence every quadratic nonresidue is also a biquadratic nonresidue. The converse also holds whenever p is a prime of the form $4n+3$.

For if, in this case, a is a quadratic residue of p, let us put $a \equiv bb \pmod{p}$. Here b is either a quadratic residue of p or a nonresidue. In the first case, let us put $b \equiv cc$, giving $a \equiv c^4$; that is, a is a biquadratic residue of p. In the second case, $-b$ is a quadratic residue of p (since -1 is a nonresidue of every prime of the form $4n + 3$). Putting $-b \equiv cc$ again gives $a \equiv c^4$, and a is a biquadratic residue of p. It is also easy to see that in this case the congruence $x^4 \equiv a \pmod{p}$ has no solutions besides $x \equiv c$ and $x \equiv -c$. Since these evident propositions exhaust the entire theory of biquadratic residues for prime moduli of the form $4n+3$, we shall exclude these moduli altogether from our investigation, restricting it to prime moduli of the form $4n + 1$.

4.

Thus, when p is a prime of the form $4n+1$, the converse of the proposition in the preceding article does not hold: there can be quadratic residues that are not also biquadratic residues. This happens whenever a quadratic residue is congruent to the square of a quadratic nonresidue. For put $a \equiv bb$, with b a quadratic nonresidue of p. If the congruence $x^4 \equiv a$ could be satisfied by a value $x \equiv c$, we would have $c^4 \equiv bb$, so that the product $(cc-b)(cc+b)$ would be divisible by p. Then p would have to divide either the factor $cc - b$ or the other factor $cc + b$; that is, either $+b$ or $-b$ would be a quadratic residue of p, and therefore both would be (since -1 is a quadratic residue), contrary to the hypothesis.

All integers not divisible by p could therefore be divided into three classes: the first would contain the biquadratic residues, the second the biquadratic nonresidues that are also quadratic residues, and the third the quadratic nonresidues. Clearly it suffices to classify only the numbers $1, 2, 3 \ldots . p - 1$. Half would belong to the third class, while the other half would be divided between the first and second classes.

5.

It will be better, however, to establish four classes as follows.

Let A be the collection of all biquadratic residues of p between 1 and $p-1$, inclusive, and let e be any chosen quadratic nonresidue of p. Further, let B be the collection of the least positive residues of the products eA modulo p, and let C, D respectively be the collections of the least positive residues of the products eeA, e^3A modulo p. It is then easy to see that the individual numbers in B are distinct, as are those in C and those in D; zero cannot occur among any of these numbers. Also, all numbers in A and C are quadratic residues of p, while all those in B and D are quadratic nonresidues. Thus neither of the collections A, C can have a number in common with B or D. But neither can A have a number in common with C, nor B with D. For suppose

I. that some number in A, say a, also occurs in C, where it arises from the congruent product eea', with a' a number in A. Put $a \equiv \alpha^4$, $a' \equiv \alpha'^4$, and choose an integer Θ such that $\Theta\alpha' \equiv 1$. We then have $ee\alpha'^4 \equiv \alpha^4$, and multiplying by Θ^4 gives

$$ee \equiv \alpha^4\Theta^4$$

That is, ee is a biquadratic residue, and hence e a quadratic residue, contrary to the hypothesis.

II. Similarly, suppose that a number belongs to both collections B, D, arising from the products ea, e^3a', with a, a' numbers in A. The congruence $ea \equiv e^3a'$ would imply $a \equiv eea'$. We would thus have a number arising from the product eea' that belonged to both C and A, which we have just proved impossible.

It is also easy to prove that *all* quadratic residues of p between 1 and $p-1$, inclusive, must occur either in A or in C, and all quadratic nonresidues of p between those limits must occur either in B or in D. For

I. Every such quadratic residue that is also a biquadratic residue belongs to A by hypothesis.

II. Write a quadratic residue h (less than p) that is also a biquadratic nonresidue as $\equiv gg$, where g is a quadratic nonresidue. Choose an integer γ such that $e\gamma \equiv g$. Then γ is a quadratic residue of p, which we write as $\equiv kk$. Hence

$$h \equiv gg \equiv ee\gamma\gamma \equiv eek^4$$

Since the least residue of k^4 belongs to A, the number h, which arises by multiplying it by ee, must belong to C.

III. Let h denote a quadratic nonresidue of p between 1 and $p-1$. Find an integer g between the same limits such that $eg \equiv h$. Then g is a quadratic

residue and so belongs to either A or C. In the first case, h evidently belongs to B, and in the second to D.

All this shows that the numbers $1, 2, 3 \ldots . p-1$ are distributed among the four lists A, B, C, D so that each occurs in one of them; each list must therefore contain $\frac{1}{4}(p-1)$ numbers. In this classification, the numbers belonging to A and C are intrinsic to those classes. The distinction between B and D, however, is arbitrary insofar as it depends on the choice of e, which itself always belongs to B. Thus, if a number from D is chosen instead, the classes B, D are interchanged.

6.

Since -1 is a quadratic residue of p, let us put $-1 \equiv ff \pmod{p}$. The four roots of the congruence $x^4 \equiv 1$ are then $1, f, -1, -f$. Thus if a is a biquadratic residue of p, say $\equiv \alpha^4$, the four roots of the congruence $x^4 \equiv a$ are $\alpha, f\alpha, -\alpha, -f\alpha$, which are easily seen to be pairwise incongruent. Hence, if we collect the least positive residues of the fourth powers $1, 16, 81, 256 \ldots . (p-1)^4$, they are always equal in groups of four, giving $\frac{1}{4}(p-1)$ distinct biquadratic residues that form the collection A. If we collect the least residues of the fourth powers only as far as $(\frac{1}{2}p - \frac{1}{2})^4$, each occurs twice.

7.

The product of two biquadratic residues is clearly a biquadratic residue: multiplying two numbers from class A always gives a product whose least positive residue belongs to the same class. Similarly, the product of a number from B and one from D, or of a number from C and one from C, has its least residue in A.

In B lie the residues of the products $A.B$ and $C.D$; in C those of $A.C$, $B.B$ and $D.D$; and finally in D those of $A.D$ and $B.C$.

The proofs are so evident that it suffices to indicate one. Let, for example, c and d be numbers from C and D, and put $c \equiv eea$, $d \equiv e^3a'$, with a, a' numbers from A. Then e^4aa' is a biquadratic residue, so its least residue belongs to A. For the product we have $cd \equiv e.e^4aa'$, so its least residue belongs to B.

We can now also easily decide which class contains a product of several factors. Assign to the classes A, B, C, D the characters $0, 1, 2, 3$, respectively. The character of a product is then either the sum of the characters of its individual factors or the least residue of that sum modulo 4.

8.

It seemed worthwhile to develop these elementary propositions without using the theory of power residues. With its help, everything can be proved much more easily still.

Let g be a primitive root for the modulus p: a number such that in the sequence of powers $g, gg, g^3 \ldots$., none before g^{p-1} is congruent to unity modulo p. Then the least positive residues of the numbers $1, g, gg, g^3 \ldots . g^{p-2}$ agree, apart from their order, with $1, 2, 3 \ldots . . p-1$, and are distributed among four classes as follows:

to	least residues of the numbers
A	$1,\ g^4, g^8, g^{12} \ldots . . g^{p-5}$
B	$g,\ g^5, g^9, g^{13} \ldots . . g^{p-4}$
C	$gg, g^6, g^{10}, g^{14} \ldots . . g^{p-3}$
D	$g^3, g^7, g^{11}, g^{15} \ldots . . g^{p-2}$

All the preceding propositions follow directly from this.

Just as the numbers $1, 2, 3 \ldots . p-1$ have here been divided among four classes, whose collections we denote by A, B, C, D, we may assign *any* integer not divisible by p to one of these classes according to its least residue modulo p.

9.

We denote by f the least residue of the power $g^{\frac{1}{4}(p-1)}$ modulo p. Since $ff \equiv g^{\frac{1}{2}(p-1)} \equiv -1$ (*Disquis. Arithm.*, Article 62), the symbol f clearly has the same meaning here as in Article 6. Thus the power $g^{\frac{1}{4}\lambda(p-1)}$, with λ a positive integer, is congruent modulo p to $1, f, -1, -f$, according as λ has the form $4m, 4m+1, 4m+2, 4m+3$, respectively, or according as the least residue of g^λ belongs to A, B, C, D, respectively. This gives us a very simple criterion for deciding which class contains a given number h not divisible by p: h belongs to A, B, C or D according as the power $h^{\frac{1}{4}(p-1)}$ is congruent modulo p to $1, f, -1$ or $-f$.

As a corollary, -1 always belongs to class A when p has the form $8n+1$, and to class C when p has the form $8n+5$. A proof of this theorem independent of the theory of power residues can easily be obtained from what we established in Article 115, III of the *Disquisitiones Arithmeticae*.

10.

Since *all* primitive roots for the modulus p arise as residues of the powers g^λ, on taking for λ all numbers coprime to $p-1$, it is easy to see that these roots are equally divided between B and D, with the base g always in B. If we replace g as base by another primitive root from B, the classification remains the same; if we use a primitive root from D as base, the classes B and D are interchanged.

If the classification is based on the criterion in the preceding article, the distinction between B and D depends on which root of the congruence $xx \equiv -1 \pmod{p}$ we choose as the characteristic number f.

11.

To make it easier to illustrate by examples the subtler investigations that we are about to undertake, we give here the construction of the classes for all moduli below 100. For each modulus we have chosen the least primitive root.

$$p = 5$$

$$g = 2,\ f = 2$$

A	1
B	2
C	4
D	3

$$p = 13$$

$$g = 2,\ f = 8$$

A	$1, 3, 9$
B	$2, 5, 6$
C	$4, 10, 12$
D	$7, 8, 11$

$$p = 17$$

$$g = 3,\ f = 13$$

A	$1, 4, 13, 16$
B	$3, 5, 12, 14$
C	$2, 8, 9, 15$
D	$6, 7, 10, 11$

$$p = 29$$

$$g = 2,\ f = 12$$

A	$1, 7, 16, 20, 23, 24, 25$
B	$2, 3, 11, 14, 17, 19, 21$
C	$4, 5, 6, 9, 13, 22, 28$
D	$8, 10, 12, 15, 18, 26, 27$

$$p = 37$$

$$g = 2,\ f = 31$$

A	$1, 7, 9, 10, 12, 16, 26, 33, 34$
B	$2, 14, 15, 18, 20, 24, 29, 31, 32$
C	$3, 4, 11, 21, 25, 27, 28, 30, 36$
D	$5, 6, 8, 13, 17, 19, 22, 23, 35$

$$p = 41$$

$$g = 6,\ \ f = 32$$

A	$1, 4, 10, 16, 18, 23, 25, 31, 37, 40$
B	$6, 14, 15, 17, 19, 22, 24, 26, 27, 35$
C	$2, 5, 8, 9, 20, 21, 32, 33, 36, 39$
D	$3, 7, 11, 12, 13, 28, 29, 30, 34, 38$

$$p = 53$$

$$g = 2,\ \ f = 30$$

A	$1, 10, 13, 15, 16, 24, 28, 36, 42, 44, 46, 47, 49$
B	$2, 3, 19, 20, 26, 30, 31, 32, 35, 39, 41, 45, 48$
C	$4, 6, 7, 9, 11, 17, 25, 29, 37, 38, 40, 43, 52$
D	$5, 8, 12, 14, 18, 21, 22, 23, 27, 33, 34, 50, 51$

$$p = 61$$

$$g = 2,\ \ f = 11$$

A	$1, 9, 12, 13, 15, 16, 20, 22, 25, 34, 42, 47, 56, 57, 58$
B	$2, 7, 18, 23, 24, 26, 30, 32, 33, 40, 44, 50, 51, 53, 55$
C	$3, 4, 5, 14, 19, 27, 36, 39, 41, 45, 46, 48, 49, 52, 60$
D	$6, 8, 10, 11, 17, 21, 28, 29, 31, 35, 37, 38, 43, 54, 59$

$$p = 73$$

$$g = 5,\ \ f = 27$$

A	$1, 2, 4, 8, 9, 16, 18, 32, 36, 37, 41, 55, 57, 64, 65, 69, 71, 72$
B	$5, 7, 10, 14, 17, 20, 28, 33, 34, 39, 40, 45, 53, 56, 59, 63, 66, 68$
C	$3, 6, 12, 19, 23, 24, 25, 27, 35, 38, 46, 48, 49, 50, 54, 61, 67, 70$
D	$11, 13, 15, 21, 22, 26, 29, 30, 31, 42, 43, 44, 47, 51, 52, 58, 60, 62$

$$p = 89$$

$$g = 3,\ \ f = 34$$

A	$1, 2, 4, 8, 11, 16, 22, 25, 32, 39, 44.\ 45,$
	$50, 57, 64, 67, 73, 78, 81, 85, 87, 88$
B	$3, 6, 7, 12, 14, 23, 24, 28, 33, 41, 43, 46,$
	$48, 56, 61, 65, 66, 75, 77, 82, 83, 86$
C	$5, 9, 10, 17, 18, 20, 21, 34, 36, 40, 42, 47,$
	$49, 53, 55, 68, 69, 71, 72, 79, 80, 84$
D	$13, 15.\ 19, 26, 27, 29.\ 30, 31, 35, 37, 38.\ 51,$
	$52, 54, 58, 59, 60, 62, 63, 70, 74, 76$

$$p = 97$$

$$g = 5,\ \ f = 22$$

A	$1, 4, 6, 9, 16, 22, 24, 33, 35, 36, 43, 47,$
	$50, 54, 61, 62, 64, 73, 75, 81, 88, 91, 93, 96$
B	$5, 13, 14, 17, 19, 20, 21, 23, 29, 30, 41, 45,$
	$52, 56, 67, 68, 74, 76, 77, 78, 80, 83, 84, 92$
C	$2, 3, 8, 11, 12, 18, 25, 27, 31, 32, 44, 48,$
	$49, 53, 65.\ 66, 70, 72, 79, 85, 86, 89, 94, 95$
D	$7, 10, 15, 26, 28, 34, 37, 38, 39.\ 40, 42, 46,$
	$51, 55, 57, 58, 59, 60, 63, 69, 71, 82, 87, 90$

12.

Since 2 is a quadratic residue of every prime of the form $8n + 1$, but a nonresidue of every prime of the form $8n + 5$, for prime moduli of the first form 2 belongs to class A or C, and for moduli of the second form to B or D. The distinction between B and D is not intrinsic, depending only on the choice of f, so we shall set moduli of the form $8n + 5$ aside for a while. An *inductive investigation* of moduli of the form $8n + 1$ gives 2 in A for $p = 73, 89, 113, 233, 257, 281, 337, 353$, etc.; in contrast, 2 belongs to C for $p = 17, 41, 97, 137, 193, 241, 313, 401, 409, 433, 449, 457$, etc.

Since, moreover, for a prime modulus of the form $8n+1$, -1 is a biquadratic residue, -2 clearly always belongs to the same class as $+2$.

13.

When we compare the examples in the preceding article, at least at first sight no simple criterion seems to offer itself for distinguishing the first moduli from the second. Nevertheless, there are *two* such criteria, remarkable for their elegance and simplicity. The following considerations will prepare the way to one of them.

The modulus p, being a prime of the form $8n+1$, can be expressed in the form $aa+2bb$ in exactly one way (*Disquiss. Arithm.*, Article 182, II). We assume that the roots a, b are taken positive. Clearly a is odd and b even; let us put $b = 2^\lambda c$, with c odd. We now observe:

I. Since $p \equiv aa \pmod{c}$, p is a quadratic residue of c, and hence also of each prime factor into which c is resolved. Conversely, by the fundamental theorem, each of these prime factors is a quadratic residue of p, and so their product c is also a quadratic residue of p. Since this also holds for 2, b is clearly a quadratic residue of p, and hence both bb and $-bb$ are biquadratic residues.

II. Thus $-2bb$ must belong to the same class as 2. Since $aa \equiv -2bb$, 2 clearly belongs to class A or C according as a is a quadratic residue of p or a quadratic nonresidue.

III. Now suppose that a has been resolved into prime factors. Denote those of the form $8m+1$ or $8m+7$ by $\alpha, \alpha', \alpha''$, etc., and those of the form $8m+3$ or $8m+5$ by β, β', β'', etc.; let the number of the latter be $= \mu$. Since $p \equiv 2bb \pmod{a}$, p is a quadratic residue of those prime factors of a of which 2 is a quadratic residue, that is, of $\alpha, \alpha', \alpha''$, etc.; it is a quadratic nonresidue of those factors of which 2 is a quadratic nonresidue, that is, of β, β', β'', etc. Conversely, by the fundamental theorem, each of $\alpha, \alpha', \alpha''$, etc., is a quadratic residue of p, while each of β, β', β'', etc., is a quadratic nonresidue. We therefore conclude that their product a is a quadratic residue of p or a nonresidue according as μ is even or odd.

IV. It is easy to verify that the product of all $\alpha, \alpha', \alpha''$, etc., has the form $8m+1$ or $8m+7$, and that the same holds for the product of all β, β', β'', etc., when their number is even. In this case the product a must therefore also have the form $8m+1$ or $8m+7$. In contrast, the product of all β, β', β'', etc., when their number is odd, has the form $8m+3$ or $8m+5$, and the same therefore holds for the product a in this case.

From all this we obtain the elegant theorem:

Whenever a has the form $8m+1$ or $8m+7$, 2 belongs to A; whenever a has the form $8m+3$ or $8m+5$, 2 belongs to C.

The examples listed in the preceding article confirm this. The first moduli decompose as follows: $73 = 1 + 2 \,.\, 36$, $89 = 81 + 2 \,.\, 4$, $113 = 81 + 2 \,.\, 16$, $233 = 225 + 2 \,.\, 4$, $257 = 225 + 2 \,.\, 16$, $281 = 81 + 2 \,.\, 100$, $337 = 49 + 2 \,.\, 144$, $353 = 225 + 2 \,.\, 64$; the second as follows: $17 = 9 + 2 \,.\, 4$, $41 = 9 + 2 \,.\, 16$, $97 = 25 + 2 \,.\, 36$, $137 = 9 + 2 \,.\, 64$, $193 = 121 + 2 \,.\, 36$, $241 = 169 + 2 \,.\, 36$, $313 = 25 + 2 \,.\, 144$, $401 = 9 + 2 \,.\, 196$, $409 = 121 + 2 \,.\, 144$, $433 = 361 + 2 \,.\, 36$, $449 = 441 + 2 \,.\, 4$, $457 = 169 + 2 \,.\, 144$.

14.

The decomposition of p into a square and twice a square has revealed such a remarkable connection with the classification of 2 that it seems worthwhile to ask whether a decomposition into two squares, which p is also known to admit, might give a similar result. Here are the decompositions of the numbers p for which 2 belongs to class

A	C
$9 + 64$	$1 + 16$
$25 + 64$	$25 + 16$
$49 + 64$	$81 + 16$
$169 + 64$	$121 + 16$
$1 + 256$	$49 + 144$
$25 + 256$	$225 + 16$
$81 + 256$	$169 + 144$
$289 + 64$	$1 + 400$
	$9 + 400$
	$289 + 144$
	$49 + 400$
	$441 + 16$

First we observe that, of the two squares into which p decomposes, one must be odd, which we put $= aa$, and the other even, which we put $= bb$. Since aa has the form $8n+1$, values of b that are twice an odd number clearly correspond to values of p of the form $8n + 5$, which are excluded from our inductive investigation here, since they would have 2 in class B or D. For values of p of the form $8n + 1$, however, b must be divisible by four. If we may trust the inductive evidence presented by this table, 2 belongs to class A for every modulus for which b has the form $8n$, and to class C for every modulus for which b has the form $8n + 4$. This theorem, however, requires much deeper investigation than the one obtained in the preceding article. Its proof must be preceded by several preliminary investigations concerning the order in which the numbers in A, B, C, D follow one another.

15.

Let us denote the number of numbers in A immediately followed by a number in A, B, C, D, respectively, by $(00), (01), (02), (03)$; likewise the number of numbers in B followed by a number in A, B, C, D, respectively, by $(10), (11), (12), (13)$. Similarly, let C contain respectively $(20), (21), (22), (23)$ numbers, and D contain $(30), (31), (32), (33)$ numbers, followed by a number in A, B, C, D. We propose to determine these sixteen counts a priori. To help readers compare the general reasoning with examples, it seemed useful to give here the numerical values of the entries in the scheme (S)

$$\begin{array}{llll} (00), & (01), & (02), & (03) \\ (10), & (11), & (12), & (13) \\ (20). & (21), & (22), & (23) \\ (30), & (31), & (32), & (33) \end{array}$$

for each modulus whose classification we gave in Article 11.

$p = 5$	$p = 13$	$p = 17$	$p = 29$
$0, 1, 0.\ 0$	$0, 1, 2, 0$	$0, 2, 1, 0$	$2.\ 3, 0.\ 2$
$0, 0, 0, 1$	$1, 1, 0, 1$	$2, 0, 1, 1$	$1, 1, 2, 3$
$0, 0, 0, 0$	$0, 1, 0, 1$	$1, 1, 1, 1$	$2, 1, 2.\ 1$
$0, 0, 1, 0$	$1, 0, 1, 1$	$0, 1, 1, 2$	$1, 2, 3, 1$

$p = 37$	$p = 41$	$p = 53$	$p = 61$
$2, 1, 2, 4$	$0, 4, 3, 2$	$2, 3, 6, 2$	$4, 3, 2, 6$
$2, 2, 4, 1$	$4, 2, 2, 2$	$4, 4, 2, 3$	$3, 3, 6, 3$
$2, 2, 2, 2$	$3, 2, 3, 2$	$2, 4, 2, 4$	$4, 3, 4, 3$
$2, 4, 1, 2$	$2, 2, 2, 4$	$4, 2, 3, 4$	$3, 6, 3, 3$

$p = 73$	$p = 89$	$p = 97$
$5, 6, 4, 2$	$3, 8, 6, 4$	$2, 6, 7, 8$
$6, 2, 5, 5$	$8, 4, 5, 5$	$6, 8, 5, 5$
$4, 5, 4, 5$	$6, 5, 6, 5$	$7, 5, 7, 5$
$2, 5, 5, 6$	$4, 5, 5, 8$	$8, 5, 5, 6$

Since moduli of the forms $8n + 1$ and $8n + 5$ behave differently, we must treat them separately. We begin with the first.

16.

The symbol (00) indicates how many different ways the equation $\alpha + 1 = \alpha'$ can be satisfied, with α, α' denoting arbitrary numbers in A. For a modulus of the form $8n + 1$, as understood here, α' and $p - \alpha'$ belong to the same

collection. We may therefore say more elegantly that (00) expresses the number of different ways of satisfying the equation $1 + \alpha + \alpha' = p$. Clearly the congruence $1 + \alpha + \alpha' \equiv 0 \pmod{p}$ can also take the place of this equation.

Likewise,

(01) indicates the number of solutions of the congruence $1+\alpha+\beta \equiv 0 \pmod{p}$
(02) the number of solutions of the congruence $1 + \alpha + \gamma \equiv 0$
(03) the number of solutions of the congruence $1 + \alpha + \delta \equiv 0$
(11) the number of solutions of the congruence $1 + \beta + \beta' \equiv 0$, etc.

where β and β' denote arbitrary numbers in B, γ numbers in C, and δ numbers in D. We immediately obtain the following six equations:

$$(01) = (10), \quad (02) = (20), \quad (03) = (30),$$
$$(12) = (21), \quad (13) = (31), \quad (23) = (32)$$

From any given solution of the congruence $1 + \alpha + \beta \equiv 0$ we obtain a solution of $1 + \delta + \delta' \equiv 0$ by taking for δ the number between $1 \ldots . p - 1$ that satisfies $\beta\delta \equiv 1$ (which clearly belongs to D), and for δ' the least positive residue of the product $\alpha\delta$ (which likewise belongs to D). The reverse passage from a given solution of $1 + \delta + \delta' \equiv 0$ to a solution of $1 + \alpha + \beta \equiv 0$ is equally clear: choose β so that $\beta\delta \equiv 1$, and at the same time put $\alpha \equiv \beta\delta'$. Thus the two congruences have the same number of solutions, or $(01) = (33)$.

Similarly, from the congruence $1 + \alpha + \gamma \equiv 0$ we derive $\gamma' + \gamma'' + 1 \equiv 0$ by choosing γ' in C so that $\gamma\gamma' \equiv 1$, and γ'' in the same collection congruent to the product $\alpha\gamma'$. It follows easily that these two congruences admit the same number of solutions, or $(02) = (22)$.

Likewise, from $1 + \alpha + \delta \equiv 0$ we derive $\beta + \beta' + 1 \equiv 0$ by choosing β, β' so that $\beta\delta \equiv 1$, $\beta\alpha \equiv \beta'$, giving $(03) = (11)$.

Finally, from $1 + \beta + \gamma \equiv 0$ we similarly derive both $\delta + 1 + \beta' \equiv 0$ and $\gamma' + \delta' + 1 \equiv 0$, and conclude that $(12) = (13) = (23)$.

We have thus obtained eleven equations among our sixteen unknowns, reducing them to five. The scheme S can therefore be written as

$$\begin{array}{cccc} h, & i, & k, & l \\ i, & l, & m, & m \\ k, & m, & k, & m \\ l, & m, & m, & i \end{array}$$

Three further equations of condition can easily be added. Every number in A, except the last, $p - 1$, must be followed by a number in one of the collections A, B, C or D. Thus

$$(00) + (01) + (02) + (03) = 2n - 1$$

and likewise

$$(10)+(11)+(12)+(13)=2n$$
$$(20)+(21)+(22)+(23)=2n$$
$$(30)+(31)+(32)+(33)=2n$$

In the notation just introduced, the first three equations give

$$h+i+k+l=2n-1$$
$$i+l+2m=2n$$
$$k+m=n$$

The fourth is identical to the second. These equations allow us to eliminate three unknowns, so that all sixteen have now been reduced to two.

17.

To determine them completely, it will be useful to investigate the number of solutions of the congruence

$$1+\alpha+\beta+\gamma\equiv 0 \pmod{p}$$

where α,β,γ denote arbitrary numbers in A,B,C. Clearly $\alpha=p-1$ is inadmissible, since $\beta+\gamma\equiv 0$ cannot hold. Substituting the remaining values for α in turn gives h,i,k,l values of $1+\alpha$ belonging to A,B,C,D, respectively. For each *given* value of $1+\alpha$ belonging to A, say $1+\alpha=\alpha^0$, the congruence $\alpha^0+\beta+\gamma\equiv 0$ admits as many solutions as $1+\beta'+\gamma'\equiv 0$ (put $\beta\equiv\alpha^0\beta'$, $\gamma\equiv\alpha^0\gamma'$), that is, $(12)=m$ solutions. Likewise, for each given value of $1+\alpha$ belonging to B, say $1+\alpha=\beta^0$, the congruence $\beta^0+\beta+\gamma\equiv 0$ has as many solutions as $1+\alpha'+\beta'\equiv 0$ (put $\beta\equiv\beta^0\alpha'$, $\gamma\equiv\beta^0\beta'$), that is, $(01)=i$ solutions. Similarly, for each given value of $1+\alpha$ belonging to C, say $1+\alpha=\gamma^0$, the congruence $\gamma^0+\beta+\gamma\equiv 0$ can be solved in as many different ways as $1+\delta+\alpha'\equiv 0$ (put $\beta\equiv\gamma^0\delta$, $\gamma\equiv\gamma^0\alpha'$), so the number of solutions is $(03)=l$. Finally, for each given value of $1+\alpha$ belonging to D, say $1+\alpha=\delta^0$, the congruence $\delta^0+\beta+\gamma\equiv 0$ has as many solutions as $1+\gamma'+\delta'\equiv 0$ (put $\beta\equiv\delta^0\gamma'$, $\gamma\equiv\delta^0\delta'$), that is, $(23)=m$ solutions. Adding all these counts, the congruence $1+\alpha+\beta+\gamma\equiv 0$ clearly admits

$$hm+ii+kl+lm$$

different solutions.

In exactly the same way we find that, on substituting for β the numbers in B in turn, the sum $1+\beta$ takes respectively $(10),(11),(12),(13)$, or i,l,m,m, values belonging to A,B,C,D. For each *given* value of $1+\beta$ belonging to these collections, the congruence $1+\beta+\alpha+\gamma\equiv 0$ admits respectively

$(02), (31), (20), (13)$, or k, m, k, m, different solutions. The total number of solutions is therefore

$$= ik + lm + km + mm$$

We obtain the same value by basing the calculation on the values of the sum $1 + \gamma$.

18.

These two expressions for the same count give the equation

$$0 = hm + ii + kl - ik - km - mm$$

and hence, eliminating h by means of $h = 2m - k - 1$,

$$0 = (k - m)^2 + ii + kl - ik - kk - m$$

The last two equations of Article 16 give $k = \frac{1}{2}(l + i)$. Substituting this value turns $ii + kl - ik - kk$ into $\frac{1}{4}(l - i)^2$, so the preceding equation, multiplied by 4, becomes

$$0 = 4(k - m)^2 + (l - i)^2 - 4m$$

Since $4m = 2(k + m) - 2(k - m) = 2n - 2(k - m)$, it follows that

$$2n = 4(k - m)^2 + 2(k - m) + (l - i)^2$$

or

$$8n + 1 = (4(k - m) + 1)^2 + 4(l - i)^2$$

Thus, putting

$$4(k - m) + 1 = a, \quad 2l - 2i = b$$

we have

$$p = aa + bb$$

It is known that p can be decomposed into two squares in exactly one way, with the odd square taken as aa and the even square as bb, so aa, bb are completely determined numbers. But a itself is also completely determined: the square root must be taken positive or negative according as the positive root has the form $4M+1$ or $4M+3$. We shall shortly discuss how to determine the sign of b.

Combining these new equations with the last three equations of Article 16 completely determines the five numbers h, i, k, l, m in terms of a, b and n, as follows:

$$\begin{aligned} 8h &= 4n - 3a - 5 \\ 8i &= 4n + a - 2b - 1 \\ 8k &= 4n + a - 1 \\ 8l &= 4n + a + 2b - 1 \\ 8m &= 4n - a + 1 \end{aligned}$$

If, in place of n, we prefer to use the modulus p, the scheme S, with every entry multiplied by 16 to avoid fractions, is

$p-6a-11$	$p+2a-4b-3$	$p+2a-3$	$p+2a+4b-3$
$p+2a-4b-3$	$p+2a+4b-3$	$p-2a+1$	$p-2a+1$
$p+2a-3$	$p-2a+1$	$p+2a-3$	$p-2a+1$
$p+2a+4b-3$	$p-2a+1$	$p-2a+1$	$p+2a-4b-3$

19.

It remains to show how to choose the sign of b. In Article 10 we pointed out that the distinction between B and D, which is not intrinsic, depends on the choice of f, for which either root of $xx \equiv -1$ must be taken, and that the two collections are interchanged when one root is replaced by the other. Inspection of the scheme just given shows that a similar interchange accompanies a change in the sign of b. We can therefore foresee that there must be a connection between the sign of b and f. To find it, we first observe that, if μ denotes a nonnegative integer and z takes all the values $1, 2, 3 \ldots . p-1$, then modulo p we have either $\Sigma z^\mu \equiv 0$ or $\Sigma z^\mu \equiv -1$, according as μ is not divisible by $p-1$ or is divisible by it. The latter part of the theorem follows from the fact that for a value of μ divisible by $p-1$, $z^\mu \equiv 1$. We prove the former part as follows. Let g denote a primitive root. All the values of z coincide with the least residues of all g^y when y takes all the values $0, 1, 2, 3 \ldots . p-2$, and hence $\Sigma z^\mu \equiv \Sigma g^{\mu . y}$. Now

$$\Sigma g^{\mu y} = \frac{g^{\mu(p-1)} - 1}{g^\mu - 1}, \quad \text{and hence}$$
$$(g^\mu - 1)\Sigma z^\mu \equiv g^{\mu(p-1)} - 1 \equiv 0$$

For a value of μ not divisible by $p-1$, g^μ cannot be congruent to 1, or equivalently $g^\mu - 1$ cannot be divisible by p. Hence $\Sigma z^\mu \equiv 0$. Q. E. D.

Now expand the power $(z^4+1)^{\frac{1}{4}(p-1)}$ by the binomial theorem. The preceding lemma gives

$$\Sigma(z^4+1)^{\frac{1}{4}(p-1)} \equiv -2 \text{ (mod. } p)$$

The least residues of all z^4 give all the numbers in A, each occurring four times. Among the least residues of z^4+1 we therefore have

$$\begin{array}{ll} 4(00) & \text{to } A \\ 4(01) & \text{to } B \\ 4(02) & \text{to } C \\ 4(03) & \text{to } D \end{array}$$

belonging to those classes, and four are $= 0$ (namely, when $z^4 \equiv p-1$). Using the criteria for the collections A, B, C, D, we obtain

$$\begin{aligned} \Sigma(z^4+1)^{\frac{1}{4}(p-1)} \equiv\ & 4(00) + 4f(01) \\ & - 4(02) - 4f(03) \end{aligned}$$

and hence

$$-2 \equiv 4(00) + 4f(01) - 4(02) - 4f(03)$$

or, substituting for $(00), (01)$, etc., the values found in the preceding article,

$$-2 \equiv -2a - 2 - 2bf$$

We therefore conclude that we must always have $a + bf \equiv 0$, or, multiplying by f,

$$b \equiv af$$

This congruence determines the sign of b when f has already been chosen, or determines f when the sign of b is prescribed in some other way.